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# Undergraduate Graduation Project

(Academic Thesis)

**Title:** CT-MUSIQ: An Automated Perceptual Quality A

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# Abstract

**Computer Science and Technology**

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Low-Dose Computed Tomography (LDCT) has become a cornerstone of modern clinical diagnostics, enabling substantial reductions in patient radiation exposure without sacrificing diagnostic utility. However, the inherent trade-off between dose reduction and image quality introduces complex, non-stationary noise degradations and streak artefacts that critically impair radiological assessment. Accurate, automated evaluation of perceptual image quality is therefore essential for quality assurance in high-throughput clinical pipelines, yet the field lacks reliable, no-reference methods specifically designed for medical CT imaging.

This thesis proposes **CT-MUSIQ**, a novel no-reference image quality assessment (NR-IQA) framework that adapts the Multi-Scale Image Quality Transformer (MUSIQ) architecture to the specific biophysical and perceptual constraints of brain CT imaging. CT-MUSIQ accepts a single axial CT slice as input, applies the clinical brain soft-tissue windowing transformation (level 40 HU, width 80 HU), and produces a continuous perceptual quality score in the range $[0, 4]$ that closely aligns with consensus radiologist Likert scale annotations.

The proposed architecture introduces four key innovations over the vanilla MUSIQ baseline: (1) a **multi-scale spatial pyramid** operating at two resolutions ($224 \times 224$ and $384 \times 384$ pixels), producing 193 patch tokens per image that simultaneously capture global structural context and fine-grained noise texture; (2) a **hash-based positional encoding** scheme that injects scale, row, and column identity into each token without the sequence-length restrictions of standard sinusoidal encodings; (3) a **scale-consistency Kullback–Leibler divergence regularisation** loss that penalises disagreement between per-scale quality predictions and the global head, preventing the model from learning scale-specific shortcuts on the small training dataset; and (4) a **pairwise margin ranking loss** component that explicitly optimises the rank-based correlation metrics used in the LDCTIQAC 2023 grand challenge evaluation protocol.

The system is trained on the LDCTIQAC 2023 dataset comprising 1,000 brain CT slices annotated by expert radiologists, using a two-stage fine-tuning strategy with ViT-B/32 pretraining, mixed-precision training, exponential moving average weight averaging, and gradient clipping, designed to fit within 6 GB of consumer GPU memory.

Experimental evaluation on the held-out test set of 100 images demonstrates that CT-MUSIQ

achieves a Pearson Linear Correlation Coefficient (PLCC) of **0.9498**, a Spearman Rank-Order Correlation Coefficient (SROCC) of **0.9488**, a Kendall Rank-Order Correlation Coefficient (KROCC) of **0.8249**, and a composite Aggregate score of **2.7235**, placing it second among all methods evaluated relative to the published LDCTIQAC 2023 leaderboard. An ensemble baseline (Agaldran Combo: Swin-T + ResNet-50) achieves an Aggregate of 2.7682, outperforming the published leaderboard top submission of 2.7427, with CT-MUSIQ remaining highly competitive. Ablation studies confirm the individual contributions of each architectural component, with the scale-consistency KL loss providing the most significant regularisation benefit on the limited training data.

These results demonstrate that a carefully adapted Vision Transformer architecture with domain-specific windowing, multi-scale tokenisation, and scale-consistency regularisation can achieve state-of-the-art automated perceptual quality assessment for brain CT imaging, providing a computationally efficient and clinically interpretable foundation for integration into real-world Picture Archiving and Communication Systems (PACS) pipelines.

**Keywords:** Computed Tomography, Low-Dose CT, No-Reference Image Quality Assessment, Vision Transformer, MUSIQ, Perceptual Quality, Brain CT, Multi-Scale Learning, KL Divergence Regularisation.

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# 1 Introduction

## 1.1 Background and Motivation

Computed Tomography (CT) has established itself as one of the most indispensable modalities in modern clinical medicine. Since its clinical introduction by Hounsfield and Cormack in the early 1970s, CT imaging has undergone decades of technological evolution, culminating in today's multi-detector row scanners capable of acquiring full-body volumetric datasets in under ten seconds [1]. The exceptional anatomical resolution, rapid acquisition time, and wide clinical accessibility of CT have made it an irreplaceable tool for emergency triage, oncological staging, neurological assessment, and pre-surgical planning. Globally, hundreds of millions of CT scans are performed annually, generating an enormous volume of cross-sectional healthcare imaging data.

Despite its clinical indispensability, CT imaging carries an inherent risk that has long concerned the medical physics and radiology communities: exposure to ionising X-ray radiation. Unlike natural background radiation, medical X-ray exposure is controllable, and the clinical community has adopted the ALARA principle (As Low As Reasonably Achievable) as the governing framework for dose management [2]. Under this principle, Low-Dose CT (LDCT) protocols have been aggressively developed and deployed, achieving dose reductions of 50–80% compared to conventional normal-dose CT (NDCT) without compromising diagnostic yield for a broad range of indications. The clinical validation of LDCT has been particularly compelling in the context of lung cancer screening programmes, where multi-institutional observational studies have demonstrated that ultra-low-dose protocols can maintain the sensitivity necessary for early-stage nodule detection [3].

However, dose reduction is not without consequence. The statistical nature of X-ray detection imposes a fundamental physical trade-off: reducing the number of X-ray photons incident on the detector array directly degrades the signal-to-noise ratio of the acquired sinogram. Specifically, the photon count at each detector cell follows a Poisson distribution, and the variance of this distribution scales linearly with the mean photon count. When the mean count is reduced, the relative noise (the coefficient of variation) increases as the inverse square root of the count, producing a complex, spatially non-stationary noise field in the reconstructed image. Superimposed on the quantum noise is the Gaussian-distributed electronic readout noise of the data acquisition system, creating the well-known "Poisson-plus-Gaussian" degradation model of LDCT [4]. In the image domain, this manifests as conspicuous streak artefacts, grainy noise texture, loss of low-contrast lesion conspicuity, and blurring of subtle anatomical

boundaries.

The clinical impact of these degradations is not uniform across tissues or disease states. In brain CT, the primary diagnostic challenge is to differentiate structures with inherently small HU differences: grey matter ($\sim$40 HU), white matter ($\sim$35 HU), cerebrospinal fluid (0–15 HU), and pathological processes such as hyperacute haemorrhage (50–80 HU), oedema, and early infarction. The narrow Hounsfield unit dynamic range of brain tissue renders it exquisitely sensitive to any noise perturbation that narrows the effective contrast-to-noise ratio, making brain LDCT quality assessment a particularly demanding and clinically high-stakes problem [5].

Conventionally, CT image quality is evaluated by trained radiologists who inspect each acquired scan for diagnostic acceptability, using structured criteria covering noise level, artefact severity, contrast resolution, and overall lesion visibility. This gold standard is both the most clinically valid and the most operationally burdensome approach: it is labour-intensive, subject to significant inter-observer and intra-observer variability, and fundamentally non-scalable to the throughput demands of modern high-volume CT facilities or retrospective research analysis pipelines. As deep learning post-processing algorithms increasingly generate large numbers of reconstructed images for research evaluation or prospective quality control, the manual radiological audit bottleneck becomes acutely limiting [6].

This gap between clinical need and operational feasibility provides the primary motivation for **automated, no-reference Image Quality Assessment (NR-IQA)** for LDCT. An effective NR-IQA system would accept a single CT slice as input and predict a perceptual quality score aligned with radiologist assessment, enabling:

- Real-time scan rejection or flagging in the scanner room, prompting immediate re-acquisition at higher dose if a scan is below diagnostic threshold;
- Large-scale retrospective quality auditing of CT archives without radiologist involvement;
- Objective, reproducible benchmarking of competing LDCT denoising and reconstruction algorithms;
- Integration into clinical Picture Archiving and Communication Systems (PACS) as an automated quality gate before image distribution to radiologists.

Achieving these goals requires overcoming a series of fundamental technical challenges, which have resisted satisfactory resolution by prior computational approaches, as reviewed in detail in Chapter 2.

## 1.2 Problem Statement

The formal task addressed in this thesis is defined as follows. Given a single axial brain CT slice $\mathbf{x} \in \mathbb{R}^{H \times W}$ (with $H = W = 512$) acquired under low-dose protocol, the objective is to learn a function:

$$f_\theta : \mathbb{R}^{H \times W} \to [0,4] \tag{1.1}$$

that maps the input CT image to a continuous quality score $\hat{q} = f_\theta(\mathbf{x})$ maximally correlated with the consensus perceptual quality score $q \in [0,4]$ assigned by a panel of expert radiologists using a five-point Likert scale. Crucially, $f_\theta$ operates in the **no-reference** (or blind) setting: no paired high-quality reference image is available either during training inference, reflecting the clinical reality that acquiring such reference images would itself require an additional radiation dose.

The specific challenges that distinguish this problem from standard NR-IQA on natural photographs include:

1. **Domain specificity of degradations.** CT image noise arises from physical quantum mechanics and detector electronics rather than from digital compression or optical blur. The resulting non-stationary Poisson-driven noise does not conform to the Natural Scene Statistics (NSS) assumptions underlying classical NR-IQA algorithms [7, 8].
2. **Calibrated Hounsfield Unit scale.** Unlike RGB natural images, CT pixel values are physical quantities with precise clinical meaning. The clinically relevant intensity range for brain tissue spans only 80 HU (0 to 80 HU after brain soft-tissue windowing), meaning that the specific preprocessing applied to the image before assessment fundamentally shapes the perceptual quality distribution seen by both the human radiologist and any automated model.
3. **Global artefact structure.** LDCT streak artefacts arising from photon starvation propagate across the entire image cross-section along Radon transform directions, creating long-range spatial correlations that local convolutional kernels cannot model with finite receptive fields.
4. **Extremely limited annotated training data.** Radiologist annotation of CT images is expensive and ethically constrained. The largest publicly available annotated LDCT IQA dataset, released by the LDCTIQAC 2023 challenge, contains only 1,000 labelled images. Training deep neural networks of the scale required for state-of-the-art IQA on such limited data demands rigorous regularisation strategies.
5. **Input resolution paradox.** High-resolution CT images (512 $\times$ 512 pixels) must be processed to capture fine-grained noise signatures, yet standard Vision Transformer architectures require fixed-resolution inputs and cannot natively process variable-length

patch sequences across multiple scales simultaneously.

## 1.3 Proposed Approach: CT-MUSIQ

This thesis proposes **CT-MUSIQ**, a purpose-built NR-IQA framework that addresses each of the above challenges through a cohesive set of architectural and training innovations. CT-MUSIQ is constructed as a principled medical domain adaptation of the Multi-Scale Image Quality Transformer (MUSIQ) [9], an architecture originally designed for natural image quality assessment that was selected as the foundation for this work due to its unique ability to process images at multiple resolutions without interpolation-induced information loss.

The core design philosophy of CT-MUSIQ rests on three guiding principles:

**Principle 1: Clinical fidelity through domain-specific preprocessing.** Rather than applying standard ImageNet normalisation to CT images, CT-MUSIQ enforces the brain soft-tissue CT window transformation (level 40 HU, width 80 HU) that replicates the exact intensity range presented to annotating radiologists during the labelling process. This ensures that the model learns quality representations in the same perceptual space as the human ground truth labels.

**Principle 2: Multi-scale global receptive field from the first layer.** By extracting non-overlapping $32 \times 32$ patches from the CT image at both $224 \times 224$ and $384 \times 384$ resolutions, CT-MUSIQ simultaneously provides the Transformer encoder with coarse-scale global structural context (49 patches per image) and fine-scale local noise texture (144 patches per image), without any spatial downsampling that would destroy frequency information critical for noise assessment.

**Principle 3: Regularisation through scale consistency.** Given the small training set, the model must be prevented from learning scale-specific overfitting artefacts. The scale-consistency KL divergence loss, a novel contribution of this thesis, penalises disagreement between the quality assessments of individual scale-specific prediction heads and the global aggregated prediction, enforcing that each scale learns a coherent, internally consistent quality representation.

## 1.4 Contributions

The principal scientific and engineering contributions of this thesis are as follows:

1. **CT-MUSIQ architecture.** A complete, end-to-end trainable NR-IQA architecture adapted from MUSIQ for brain CT imaging, incorporating multi-scale patch tokenisation, hash-based positional encoding, and per-scale prediction heads. The architecture is designed to operate within 6 GB of GPU VRAM on consumer hardware using mixed-precision training.

2. **Brain CT windowing adaptation.** A principled methodological argument for and implementation of brain soft-tissue windowing as a preprocessing step that bridges the perceptual space between radiologist annotation and model training, constituting a significant departure from the abdominal windowing used in the original LDCTIQAC challenge.
3. **Scale-consistency KL regularisation.** A novel composite loss function that combines mean squared error regression, pairwise margin ranking, and Kullback–Leibler divergence between per-scale and global quality prediction distributions, verified through systematic ablation studies to provide measurable improvement over MSE-only training on the small LDCT dataset.
4. **Two-stage fine-tuning protocol.** A carefully engineered training strategy that first warms up domain-specific components with the Transformer encoder frozen, then jointly fine-tunes all parameters with cosine-annealed learning rate decay and exponential moving average weight smoothing, achieving stable convergence on 700 training samples.
5. **Comprehensive empirical evaluation.** A full benchmark of CT-MUSIQ against both published LDCTIQAC 2023 leaderboard methods and a purpose-built strong baseline (Swin-T + ResNet-50 ensemble), with ablation studies confirming the individual contribution of each proposed component, and an analysis of the complementary strengths of the two approaches.
6. **Open-source reproducible implementation.** A complete Python/PyTorch implementation of CT-MUSIQ, including all training, evaluation, and ablation scripts, centralised configuration management, and deterministic reproducibility through fixed random seeds, released as a fully documented open-source repository.

## 1.5 Thesis Structure

The remainder of this thesis is organised as follows.

**Chapter 2: Literature Review** surveys the relevant prior work across five thematic domains: the clinical evolution of CT reconstruction algorithms; the theoretical and practical limitations of full-reference IQA in the medical imaging context; the failure modes of classical NSS-based NR-IQA when applied to CT data; the development of deep learning-based NR-IQA methods including CNN-based and Transformer-based architectures; and the emergence of vision-language models as the current frontier of IQA research. The chapter concludes with a precise formulation of the research gap addressed by this thesis.

**Chapter 3: Methodology** provides a comprehensive technical description of all components of CT-MUSIQ: the dataset and preprocessing pipeline, the model architecture in full mathematical detail, the composite loss function, the two-stage training procedure, data

augmentation strategy, evaluation protocol, and the comparison baseline architecture.

**Chapter 4: Experiments** describes the experimental design including implementation details, hardware specifications, training hyperparameters, and the complete ablation study protocol. The design of each ablation configuration is motivated by the specific architectural hypotheses it tests.

**Chapter 5: Results and Discussion** presents the quantitative performance of CT-MUSIQ on the LDCTIQAC 2023 test set relative to all published leaderboard methods and the Agaldran Combo baseline, followed by detailed analysis of the ablation study results, examination of correlation metric components, and discussion of the implications of the findings for clinical deployment.

**Chapter 6: Conclusion and Future Work** synthesises the key contributions and findings of this thesis, critically examines the limitations of the proposed approach, and outlines a roadmap of promising directions for future research in automated CT quality assessment.

**Appendices** provide supplementary material including the complete hyperparameter summary table, training curve data, algorithm pseudocode, and implementation notes for reproducing all experiments.

## 1.6 Scope and Limitations

This thesis focuses specifically on axial brain CT slices from the LDCTIQAC 2023 dataset and does not claim generalisation to other CT modalities (e.g., chest, abdominal, or orthopaedic CT), multi-slice volumetric assessment, or other medical imaging modalities. The evaluation is conducted exclusively on a single publicly available benchmark dataset; multi-institutional validation across different scanner manufacturers, acquisition protocols, and patient populations represents an important direction for future work.

The CT images used in this study are pre-processed and normalised prior to release by the challenge organisers; direct application to raw DICOM files from clinical PACS systems would require additional preprocessing steps including HU calibration verification and patient table removal.

Furthermore, the Agaldran Combo baseline used in this thesis as a within-paper comparison is not the original “agaldran” team’s submission to the LDCTIQAC 2023 challenge, but rather an independent reimplementation motivated by their published architectural choices, trained under the same conditions as CT-MUSIQ. Direct comparison with the original challenge submission is therefore approximate, though the reported leaderboard numbers are cited for reference throughout.

## 1.7 Clinical Context and Significance

To fully appreciate the significance of the automated IQA problem addressed in this thesis, it is important to understand the clinical workflow in which CT image quality assessment occurs and the specific consequences of quality failures in routine clinical practice.

### 1.7.1 The CT Acquisition Workflow

In a typical clinical CT examination, the following sequence of events occurs: (1) a clinician orders a CT scan for a patient based on clinical indications; (2) the patient is positioned on the CT scanner table and the acquisition parameters (kVp, mAs, scan duration) are set by the radiographer according to local protocol, often using dose-modulation algorithms; (3) the scanner acquires the raw sinogram data and performs on-line reconstruction using the configured algorithm (FBP, IR, or DLIR); (4) the reconstructed image series is transferred to the hospital PACS; (5) a radiologist accesses the images from PACS, applies windowing, and performs a diagnostic assessment; (6) a radiological report is generated and communicated to the referring clinician.

In this workflow, quality assessment traditionally occurs implicitly at step (5): the radiologist observes the image quality as part of the diagnostic reading and may note quality issues in the report or, in severe cases, request a repeat scan. However, by step (5), the patient has already left the scanner room, and a repeat scan requires re-booking the patient, additional radiation exposure, additional scanner time, and delay in diagnosis. An automated quality gate at step (3) or (4)—before the patient leaves the room—would enable immediate repeat acquisition when necessary, avoiding all of these downstream costs.

### 1.7.2 The Economic and Radiation Safety Case

The economic case for automated IQA is straightforward. A missed poor-quality scan that reaches the radiologist requires: (a) radiologist time to assess and report on a subdiagnostic image; (b) communication with the referring clinician to explain the limitation; (c) in many cases, a repeat scan booking, which incurs scanner time, staff time, and patient transport costs; and (d) a delayed diagnosis that may have clinical consequences if the pathology was urgent. The cost of a single repeat CT examination varies by region and modality but typically ranges from $200–$1,000. Even a 1% reduction in repeat scan rates for a facility performing 50,000 CT scans per year would save 500 repeat scans annually—a significant operational benefit.

From a radiation safety perspective, eliminating unnecessary repeat scans directly reduces the aggregate population radiation dose. Modern CT scanner dose modulation algorithms already minimise dose for any given diagnostic target, but a missed poor-quality scan followed by a repeat effectively doubles the patient's radiation exposure for that examination. An IQA system that prevents this scenario contributes directly to the ALARA principle that governs

clinical radiology practice.

### 1.7.3 Research Benchmarking Applications

Beyond clinical quality control, automated IQA is essential for the reproducible benchmarking of competing CT reconstruction and denoising algorithms. The literature review (Chapter 2) documents the rapidly expanding landscape of DLIR methods, each claiming performance improvements over prior art. Without an objective, perceptual quality metric aligned with radiologist judgement, comparisons between methods rely either on subjective visual inspection (which is non-reproducible and expert-dependent) or on non-perceptual metrics such as PSNR and SSIM (which often do not correlate with diagnostic quality). CT-MUSIQ provides a reproducible, scalable, radiologist-aligned quality assessment tool that can serve as an objective benchmark for evaluating LDCT reconstruction quality across large image databases.

# 2 Literature Review

## 2.1 Introduction to the Review

The problem of automated perceptual quality assessment for low-dose CT images sits at the intersection of several active research domains: CT reconstruction and denoising, image quality assessment theory and practice, deep learning with limited data, and Vision Transformer architecture design. This chapter provides a structured review of the prior art relevant to each of these domains, beginning with the clinical and technical context of CT imaging and progressing through the evolution of IQA methods from hand-crafted statistical approaches to modern neural network architectures. The chapter concludes with a precise identification of the specific gap that CT-MUSIQ is designed to address.

The review is organised into seven thematic sections: (1) the clinical imperative and evolution of CT reconstruction; (2) the dichotomy between subjective evaluation and full-reference metrics; (3) ontological flaws in traditional no-reference IQA applied to CT; (4) the deep learning revolution in NR-IQA and the feature extraction bottleneck; (5) evaluation deficits and standardisation efforts; (6) Vision Transformers, multi-scale processing, and the MUSIQ architecture; and (7) emerging frontiers in vision-language model-based IQA. The final section synthesises the identified gaps and positions the contributions of this thesis.

## 2.2 The Clinical Imperative and Evolution of Computed Tomography Reconstruction

The expansion of versatile medical imaging modalities ranging from ultrasound and magnetic resonance imaging (MRI) to positron emission tomography (PET) has fundamentally revolutionised modern clinical diagnostics, yet it is computed tomography (CT) that nowadays accounts for an enormous majority of large-scale, cross-sectional healthcare data generation [10]. While conventional normal-dose CT (NDCT) provides unparalleled, high-resolution anatomical visualisation, its diagnostic utility is fundamentally connected with ionising X-ray radiation. This radiation raises significant health concerns regarding radiation-driven genetic mutations and future cancer-related risks. Guided strictly by the ALARA (As Low As Reasonably Achievable) principle, the medical physics community has aggressively adopted Low-Dose CT (LDCT) protocols. The critical efficacy of this adaptation has been heavily validated in expansive observational studies that include trials for lung cancer screening and long-term patient tracking [11]. Dose reduction is especially important in highly sensitive demographics, such as pediatric populations undergoing ultra-low-dose abdominal scans or

cardiac CT evaluations, where optimising the signal-to-noise ratio (SNR) against radiation risk is a delicate physiological necessity [12].

Mathematically, image reconstruction is a typical inverse problem where the goal is to recover an unknown signal $\mathbf{x}$ from a corrupted measurement $\mathbf{y} = A(\mathbf{x}) + \varepsilon$, with $A$ representing the forward corruption process [13]. In LDCT, reducing the incident X-ray photons inevitably yields photon starvation at the detector array, which mathematically expands as a complex, non-stationary "Poisson + Gaussian" degradation model; the quantum noise obeys a Poisson distribution driven by photon count fluctuations, while the electronic noise of the data acquisition system is characterised by a signal-independent Gaussian variance [4]. Early analytical reconstruction methods, most notably Filtered Back Projection (FBP), dominate clinical practice due to their computational efficiency in calculating the analytical left inverse of the system matrix. However, FBP relies on idealised linear assumptions that fail catastrophically under the high-noise, photon-starved conditions of LDCT, resulting in major streak artefacts and extensive noise amplification.

To overcome these computational limitations, Model-Based Iterative Reconstruction (MBIR) and Hybrid Iterative Reconstruction (IR) algorithms were developed. These formulate the reconstruction as a highly regularised objective function combining data fidelity with complex structural priors, typically optimised using first-order methods or the Alternating Direction Method of Multipliers (ADMM) [14]. Although IR techniques successfully incorporate statistical noise modelling (e.g., Markov Random Field priors) to minimise mean squared errors, they are often computationally expensive and frequently interfere with the inherent Noise Power Spectrum (NPS) and Modulation Transfer Function (MTF) of the CT system [15]. Additionally, iterative reconstruction algorithms have been shown to drastically shift the visual texture of the images, producing a non-natural, "plastic" or "blotchy" aesthetic that significantly degrades the diagnostic conspicuity of low-contrast lesions [16].

Therefore, the industry has aggressively pivoted toward Artificial Intelligence and Deep Learning Image Reconstruction (DLIR) algorithms, such as the Residual Encoder-Decoder CNN (RED-CNN) or dual-domain frameworks, which map degraded low-dose sinograms or image patches directly to new targets, significantly outperforming IR in preserving natural noise textures and structural sharpness [17]. Recently, DLIR has evolved beyond basic feed-forward networks to embrace advanced generative approaches, including the deployment of Score-Based Generative Models (SGMs) and sophisticated diffusion predictive models like the Contextual Error-Modulated Generalised Diffusion Model (CoreDiff) [18]. Pushing the boundaries of physical modelling, state-of-the-art frameworks such as the Noise-Inspired Diffusion Model (NEED) have explicitly tuned the reverse diffusion Markov chain to mimic the specific noise characteristics of CT, utilising a modified Poisson diffusion process to align the generative pattern directly with the true degradation physics of pre-log LDCT projections

[19]. Simultaneously, at the hardware level, the development of Photon-Counting Detector CT (PCD-CT) is revolutionising multi-energy imaging by measuring individual incident X-ray photons without electronic noise interference, demanding entirely new generations of reconstruction networks [20].

The proliferation of DLIR methods, each generating large quantities of reconstructed images with varying quality characteristics, creates a pressing demand for automated, reproducible quality metrics to replace manual radiologist evaluation for benchmarking purposes. Furthermore, comparing FBP and advanced IR techniques explicitly reveals that while iterative methods drastically improve standard signal-to-noise ratios, the aggressive smoothing due to their algorithmic regularisation can reduce the diagnosis of minute pathologies compared to the unprocessed raw data [21]. This highlights why a perceptual, rather than purely signal-theoretic, quality metric is essential for clinical relevance.

## 2.3 The Dichotomy of Image Quality Assessment: Subjective Evaluation versus Full-Reference Metrics

In the highly regulated medical domain, image quality is not defined by graphical fidelity alone but rather by the measurable contribution to diagnostic accuracy. The ultimate gold standard for Image Quality Assessment (IQA) remains the subjective, diagnosis-based evaluation performed by highly trained, board-certified radiologists who assess images for critical clinical criteria, including lesion visibility, subtle structural preservation, and the suppression of hallucinated artefacts [22]. To measure the reliability of these human annotations, statistical indicators such as the Intraclass Correlation Coefficient (ICC) are extensively implemented to quantify both inter-observer agreement (consistency across different radiologists) and intra-observer reliability (repeatability of a single radiologist's assessments over time) [6].

However, subjective IQA is limited by unavoidable practical bottlenecks: it is labour-intensive, financially exorbitant, and inherently non-scalable for evaluating large groups of computational outputs. Early computational attempts to automate this assessment relied on complex mathematical model observers, such as the Channelised Hotelling Observer (CHO) and the Channelised Support Vector Machine (CSVM). These models utilise Singular Value Decomposition (SVD) and Principal Component Analysis (PCA) to perform dimensionality reduction on specific Signal-Known-Exactly/Background-Known-Exactly (SKE/BKE) detection tasks [23].

To automate the assessment of broader, generalised image degradations without task-specific requirements, objective Full-Reference IQA (FR-IQA) metrics were eagerly adopted from the computer vision domain. These algorithms compute the degradation of a test image through direct mathematical comparisons against a perfectly aligned reference counterpart, utilising metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure

(SSIM), Feature Similarity (FSIM), Visual Signal-to-Noise Ratio (VSNR), Information Fidelity Criterion (IFC), and Visual Information Fidelity (VIF) [24]. To properly demonstrate the true clinical utility of these metrics and the underlying deep learning denoising networks, researchers construct sophisticated virtual imaging simulations based directly on patient raw data; by inserting simulated lesions and noise into the projection domain, they can accurately measure the oversampled Edge Spread Function (ESF) and in-plane MTF at various clinical contrast levels (e.g., $-10$, $-20$, $-50$ HU) [25].

Despite demonstrating high mathematical correlation in strictly controlled phantom studies, FR-IQA possesses an insurmountable ontological flaw for genuine clinical application. Acquiring a perfectly aligned, artefact-free, high-dose NDCT reference image for every corresponding LDCT scan is physically, practically, and ethically impossible due to strict radiation safety constraints and the inevitability of patient motion between sequential scans [26]. This fundamental limitation provides the primary rationale for developing no-reference (NR-IQA) methods that can operate without any paired reference image.

The transition from FR-IQA to NR-IQA, while operationally necessary, introduces significant representational challenges. Without a reference, the model must infer quality degradation from the statistics of the corrupted image alone. For natural photographs, this has been achieved with considerable success by exploiting the statistical regularity of high-quality natural scenes. However, medical images do not share these statistical properties, as explored in the following section.

## 2.4 Ontological Flaws of Traditional No-Reference IQA for CT

To eliminate the strict reliance on reference data, No-Reference IQA (NR-IQA), frequently referred to as "blind IQA", was systematically developed to evaluate image quality based solely on statistical features extracted from the degraded image itself, transforming the assessment problem into a classification or regression task [27]. The most established traditional NR-IQA algorithms, such as the Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE) and the Natural Image Quality Evaluator (NIQE), operate on the foundation of Natural Scene Statistics (NSS) [28]. These algorithms extract handcrafted statistical features by calculating local Mean Subtracted Contrast Normalised (MSCN) coefficients; they rely on the fundamental assumption that high-quality, distortion-free photographic images generally exhibit a predictable, heavy-tailed Gaussian distribution, whereas distorted images deviate from this norm, allowing models to fit deviations using a Generalised Gaussian Distribution (GGD) or an Asymmetric GGD (AGGD) [29].

While this framework is mathematically elegant and effective for assessing generic photographic degradations like JPEG compression or optical blur, the direct application of NSS-based NR-IQA to medical CT results in catastrophic performance degradation [26].

Natural photographic images are shaped by broader spatial frequencies generated by reflected visible light, whereas clinical CT images represent highly specific, biophysical X-ray absorption coefficients precisely mapped to Hounsfield Units (HU). The complex, non-stationary background noise power spectrum and the spatially variant, macro-structural streak artefacts characteristic of photon-starved LDCT fundamentally violate the core assumptions of natural scene broadband models, rendering these classical NR-IQA metrics completely unreliable for predicting true diagnostic usefulness.

The LDCTIQAC 2023 challenge results corroborate this quantitatively: BRISQUE achieves an Aggregate score of only 2.1219 (PLCC $= 0.75$, SROCC $= 0.7863$, KROCC $= 0.5856$) on the leaderboard [30], demonstrating the severe underperformance of classical statistics-based approaches on calibrated medical images. The SNR-based baseline similarly achieves only 2.4026, confirming that even physically motivated signal metrics do not capture the perceptual quality criteria applied by radiologists.

Furthermore, this catastrophic breakdown of traditional statistical blind IQA is not isolated to CT. Analogous failures have been extensively documented in Magnetic Resonance (MR) imaging, where classical NR metrics are entirely incapable of accurately quantifying modality-specific degradations such as ghosting, aliasing, or spatial displacement of water and fat molecules [31]. These cross-modality failures underscore the need for dedicated, domain-adapted NR-IQA systems trained with clinical expert labels, rather than unsupervised statistical models derived from natural image databases.

## 2.5 Deep Learning-Driven NR-IQA and the Feature Extraction Bottleneck

The representational limitations of handcrafted statistical features led to the massive integration of deep learning (DL) into the NR-IQA paradigm, fundamentally transforming the field from manual statistical modelling to data-driven representation learning [32]. Early deep learning IQA solutions aggressively took advantage of pre-trained CNN backbones, such as VGG, ResNet, DenseNet, and Inception architectures, by eliminating their classification heads and utilising transfer learning to map structured, high-dimensional feature representations to continuous subjective quality scores via fully connected regression layers.

To further enhance feature extraction and prevent the loss of critical low-level visual information due to deep pooling layers, sophisticated architectures like the staircase structure were developed to systematically integrate features from intermediate network layers, ensuring the model leverages both low-level textural noise signatures and high-level semantic anatomical context simultaneously [33]. To establish explicit medical domain priors into the network's learning trajectory, researchers developed highly specialised hybrid architectures. A paramount example involves fusing Sparse Representation encoding with psychophysical Just

Noticeable Distortion (JND) models. By explicitly calculating patch-level JND maps based on luminance adaptation and texture masking gradients, these networks selectively pool features only from visually critical, distortion-sensitive anatomical regions, thereby mimicking the localised attention of the Human Visual System (HVS) [34].

Advanced deep learning frameworks have also been engineered for specific, highly complex imaging models, such as Cone-Beam CT (CBCT), where sparse-view projections are repeatedly optimised using U-Net-based restoration operators driven by time-embedding modules to suppress severe undersampling artefacts [35]. For standard LDCT, CNN-based IQA has been applied using encoder-decoder networks with transfer learning from 2D to 3D analysis, demonstrating that cross-dimensional knowledge transfer can partially address the limited data problem [36].

Despite these significant architectural improvements, pure CNN-based NR-IQA algorithms suffer from a critical geometric limitation. Standard CNNs fundamentally rely on localised convolution kernels and stacked spatial pooling to incrementally expand their receptive fields. As a direct consequence, they invariably reduce and destroy fine-grained, high-frequency quantum noise information in deeper layers, while simultaneously failing to establish the global, long-range spatial dependencies required to accurately evaluate macro-structural streak artefacts that pass through the entire anatomical cross-section. Furthermore, training these deep CNNs on diverse clinical data from multi-institutional sources introduces severe domain shift challenges, driving the necessity for advanced active learning frameworks and incremental domain adaptation strategies to effectively scale model generalisability without requiring exhaustive, dense manual delineations [37].

## 2.6 Evaluation Deficits and Standardisation

The historical progression of deep learning-based LDCT denoising and automated IQA has been severely slowed by systemic, widespread flaws in experimental evaluation protocols across the scientific literature. A comprehensive literature analysis reveals significant deficits: the implementation of insufficient objective metrics that have little to no correlation with actual clinical diagnostic value, the highly unfair selection of hyperparameters optimised specifically to negatively impact competing reference models, and an overwhelming reliance on small, homogeneous, privately held datasets that prevent robust model generalisation [38]. To comprehensively address this critical shortcoming in algorithmic standardisation and benchmarking, researchers utilise highly sophisticated physical simulations to generate low-dose estimation data from routine-dose scans; these simulations precisely model the Poisson statistics of X-ray photons and the non-Gaussian components of detector electronic noise, creating accurate datasets at precise dose reductions (e.g., 40%, 20%, and 10% simulated doses) [39].

Establishing standardised, high-fidelity datasets is not simply an academic exercise for IQA evaluation, but an absolute prerequisite for ensuring the reliability of next-generation automated medical analysis tasks, such as the precise, deep-learning-driven segmentation of COVID-19 infections and pulmonary lesions from highly degraded CT volumes [40].

The most significant advancement in standardising perceptual assessment was the **Low-Dose Computed Tomography Perceptual Image Quality Assessment Grand Challenge (LDCTIQAC)**, held in association with MICCAI 2023 and subsequently documented in Lee et al. [30]. The challenge curators released an unprecedented, massive open-source dataset comprising 1,000 abdominal CT images sourced from multi-institutional scanners, meticulously annotated by expert radiologists using a strictly calibrated 5-point Likert grading scale to embed clinical perception directly into the ground truth labels.

The challenge attracted multiple competitive teams, whose leaderboard results provide the primary benchmark for comparison in this thesis:

- **agaldran** (Aggregate = 2.7427): The top-performing submission, combining Swin Transformer and ResNet-50 backbones, with multi-head model calibration using the approach described in [41].
- **RPI_AXIS** (Aggregate = 2.6843): Second-place submission.
- **CHILL@UK** (Aggregate = 2.6719): Third-place submission.
- **FeatureNet** (Aggregate = 2.6550): Fourth-place submission.
- **SNR baseline** (Aggregate = 2.4026): Physics-motivated baseline using signal-to-noise ratio measurements.
- **BRISQUE** (Aggregate = 2.1219): Classical NSS-based blind IQA applied directly without adaptation.

To further reduce the severe risk of catastrophic overfitting when training massive deep neural networks on relatively small, specialised medical datasets, advanced regularisation and data augmentation strategies, such as the unpredictable, convex interpolation of image-label pairs known as MixUp, are being extensively investigated and integrated into state-of-the-art training pipelines to force the networks to learn robust, linear transitions between distinct quality degradation classes [41].

## 2.7 Vision Transformers, Multi-Scale Processing, and the MUSIQ Architecture

To absolutely outperform the localised mathematical inductive biases of standard CNNs, the computer vision community has recently executed a massive architectural pivot toward the Vision Transformer (ViT) approach, which visualises an image as a flattened sequence of tokenised pixel patches and employs multi-head self-attention mechanisms to dynami-

cally compute complex correlations between all tokens across the entire image [29]. This convolution-free mechanism is uniquely suited for LDCT analysis, as it establishes a global receptive field from the very first layer, enabling the network to simultaneously correlate distant, low-frequency streak artefacts and preserve overall anatomical structure without downsampling.

### 2.7.1 Vision Transformer Foundations

The foundational Vision Transformer (ViT) model, introduced by Dosovitskiy et al. in 2020, demonstrated that a pure self-attention architecture without convolutional inductive biases could achieve competitive or superior performance on large-scale image recognition benchmarks when pretrained on sufficiently large datasets. The core ViT operation tokenises an image of dimensions $H \times W$ into $N = HW/P^2$ non-overlapping patches of size $P \times P$, projects each patch to a $d$-dimensional embedding, prepends a learnable `[CLS]` token, adds position encodings, and processes the resulting sequence through $L$ Transformer encoder blocks consisting of Multi-Head Self-Attention (MHSA) and position-wise feed-forward networks (FFN).

The MHSA operation for a query matrix $Q$, key matrix $K$, and value matrix $V$ is:

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{2.1}$$

where $d_k = d/H_{\text{attn}}$ is the per-head dimension. The global attention operation allows every patch to attend to every other patch simultaneously, enabling the model to capture long-range structural dependencies that CNN architectures require many layers to approximate.

However, applying standard ViT architectures to NR-IQA introduces a significant technical barrier known as the **input resolution paradox**: rigid ViTs require fixed-resolution inputs to maintain constant sequence lengths for their learnable position embeddings, meaning native high-resolution medical slices must be globally resized. This approach fundamentally destroys the high-frequency quantum noise signatures critical for quality assessment, while rigid crops eliminate the global anatomical context necessary to evaluate macro-artefacts.

### 2.7.2 The Swin Transformer and Hierarchical Attention

The Swin Transformer introduced a hierarchical design using shifted-window attention that partially addresses the resolution challenge by computing attention within local windows and using shifted window partitioning to introduce cross-window connections. The Swin-T backbone achieves competitive performance on dense prediction tasks while maintaining linear computational complexity with respect to image size. In the context of LDCT denoising, the Swin Transformer-based Encoder-Decoder Network (STEDNet) [42] dramatically improves

noise suppression by utilising shifted-window mechanisms and skip connections for global feature fusion, constrained by a hybrid L1 and MS-SSIM loss function. Similarly, the Hybrid CNN-Transformer codec network (HCformer) [43] utilises Window/Sliding-Window Multi-head Self-Attention (W/SW-MSA) to efficiently capture information interaction across the entire image. Furthermore, sophisticated convolution-free networks like TED-net [44], multi-domain optimisation networks (MDST) that integrate Swin Transformers directly into both projection and image domain sub-networks for sparse-view CT [45], and Cross-Network Contextual (CNA) modules [46] are setting new benchmarks.

### 2.7.3 The MUSIQ Architecture

The Multi-Scale Image Quality Transformer (MUSIQ) [9], proposed by Ke et al. at ICCV 2021, represents a powerful algorithmic solution to the resolution paradox specifically in the context of NR-IQA. Rather than resizing images to a single fixed resolution, MUSIQ extracts patches from a multi-scale spatial pyramid consisting of two or more target resolutions, concatenates all patches from all scales into a single sequence, and processes this variable-length sequence through a standard Transformer encoder.

To handle the variable sequence length arising from different scales, MUSIQ introduces a **hash-based 2D spatial embedding mechanism** that encodes each patch's scale index, row index, and column index as three independent learned embedding lookups, summed to form the positional encoding:

$$\text{pos}_i = \mathbf{E}_{\text{scale}}(s_i) + \mathbf{E}_{\text{row}}(r_i) + \mathbf{E}_{\text{col}}(c_i) \tag{2.2}$$

This factored design allows MUSIQ to process images of arbitrary resolution without interpolation-induced information loss, explicitly encoding the scale identity and spatial coordinates of each patch. MUSIQ achieved state-of-the-art performance on multiple natural IQA benchmarks (KonIQ-10K [33], SPAQ, etc.) at the time of its publication, demonstrating the superiority of the multi-scale Transformer approach over CNN-based methods for quality-aware feature extraction.

### 2.7.4 Limitations of MUSIQ for Medical CT

Despite its architectural elegance and strong performance on natural image benchmarks, the vanilla MUSIQ architecture is not directly applicable to the medical CT IQA domain for several reasons:

**Natural image preprocessing assumptions.** MUSIQ normalises input images using ImageNet mean and standard deviation statistics (designed for RGB natural photographs), which is inappropriate for CT images calibrated in Hounsfield Units. The resulting mismatch be-

tween training distribution and CT pixel value distributions would cause the pretrained patch embeddings to operate outside their trained domain.

**Absence of clinical windowing.** Radiologists who annotated the LDCTIQAC dataset applied brain soft-tissue windowing before scoring. A model that does not apply identical windowing before feature extraction is processing a fundamentally different image than the one being scored, creating a systematic misalignment between the annotation and the model's perceptual input.

**No scale-consistency regularisation.** The original MUSIQ does not enforce any agreement between per-scale quality predictions, potentially allowing the model to rely on scale-specific artefacts rather than learning truly scale-invariant quality features. On a small medical dataset of 700 training samples, this risk of scale-specific overfitting is significantly elevated.

**Small dataset regime.** MUSIQ was designed for and evaluated on large natural image IQA datasets (KonIQ-10K has 10,000 images; SPAQ has 11,125 images). Adapting it to 700 training samples requires additional regularisation strategies beyond those present in the original architecture.

These limitations motivate the specific architectural adaptations introduced in CT-MUSIQ, as detailed in Chapter 3.

## 2.8 Emerging Frontiers: Vision-Language Models in IQA

Currently, the highest frontier of medical image quality assessment involves combining Large Language Models (LLMs) and Multi-modal Large Language Models (MLLMs) to replicate the cognitive, step-by-step diagnostic reasoning of clinical radiologists [47]. State-of-the-art architectures, such as IQAGPT, leverage expansive foundational pathways (e.g., PaLM, LLaMA) fine-tuned with highly specific, autoregressive language modelling targets directly on CT-IQA datasets [48]. By passing visual tokens extracted from image encoders through complex cross-modal attention mechanisms, Vision-Language Models (VLMs), including highly advanced frameworks like LLaVA [49] and InternLM-XComposer-VL, are capable of generating remarkably accurate visual descriptions and reasoning paths.

Through the implementation of strict mental and physical prompting strategies, these models seamlessly convert abstract numerical quality ratings into highly semantic, comprehensive textual descriptions (e.g., explicitly detailing the presence of "photon starvation streaks blocking the hepatic blood vessel" or "optimal preservation of low-contrast lesions in the abdominal soft-tissue window"), generating complete, explainable diagnostic quality assessment reports along with absolute numerical scores.

Architectures such as InstructBLIP [50] and CLIP-based models [51] have demonstrated strong zero-shot and few-shot transfer capabilities that are beginning to be explored for

medical imaging tasks. The incorporation of domain-specific medical knowledge into these large pre-trained models through instruction tuning represents a promising direction for reducing the dependency on large labelled medical datasets.

While these foundational VLMs offer unprecedented zero-shot explanations and represent a monumental leap toward transparent medical AI, they are currently limited by massive computational overhead, critical inference latency, and the persistent risk of hallucinating clinical metadata. Consequently, their immediate, practical deployment as low-latency, high-throughput objective quality scoring substitutes within standard, real-time hospital PACS pipelines remains highly limited, requiring further optimisation before widespread clinical adoption. Furthermore, fully understanding the underlying optimisation principles of deep transformers, specifically phenomena such as "grokking" where networks exhibit sudden, delayed leaps in generalisation performance long after achieving perfect training accuracy, remains a critical frontier in theoretical AI research [52].

## 2.9 Synthesis and Identification of the Research Gap

The comprehensive framework of the deep learning literature clearly suggests that automated NR-IQA algorithms must strictly simulate the precise perceptual characteristics of expert radiologists to be considered clinically acceptable. The following key conclusions emerge from the review:

**Classical NR-IQA is insufficient.** NSS-based methods like BRISQUE and NIQE fail catastrophically on CT images due to the mismatch between natural image statistics and the Hounsfield Unit calibration domain. LDCTIQAC challenge results provide quantitative confirmation of this failure with BRISQUE achieving only Aggregate $= 2.1219$.

**CNN-based methods are architecturally limited for CT IQA.** The localised inductive biases of convolutional networks make them poorly suited for capturing the global, long-range streak artefacts that dominate LDCT quality degradations.

**Vanilla ViT architectures face the resolution paradox.** Fixed sequence length requirements force interpolation of high-resolution CT slices, destroying the high-frequency noise signatures critical for perceptual quality assessment.

**MUSIQ resolves the resolution paradox but was not designed for CT.** The multi-scale hash-based tokenisation of MUSIQ is architecturally ideal for LDCT IQA, but the original model assumes natural RGB image statistics and lacks the clinical preprocessing and regularisation adaptations necessary for the CT domain.

**The specific research gap is:** a dedicated MUSIQ-derived architecture that enforces clinical CT windowing as a preprocessing stage, incorporates scale-consistency regularisation to prevent overfitting on small medical datasets, and is systematically evaluated against the

LDCTIQAC 2023 grand challenge benchmark.

CT-MUSIQ is proposed to directly address this gap. By synthesising cutting-edge Transformer architectures with the physical constraints of clinical neuroradiology, the CT-MUSIQ framework delivers a computationally efficient, clinically grounded, and empirically validated automated LDCT quality assessment engine. The specific architectural and training innovations that constitute CT-MUSIQ are described in full in the following chapter.

## 2.10 Transfer Learning and Domain Adaptation in Medical Imaging

A critical enabling technology for CT-MUSIQ is the transfer of representations learned on natural image datasets to the medical imaging domain. This section reviews the principles and practice of transfer learning in medical imaging contexts, with emphasis on the specific challenges and strategies relevant to quality assessment on small annotated datasets.

### 2.10.1 Fundamentals of Transfer Learning for Medical Imaging

Transfer learning in deep neural networks leverages the observation that features learned by networks trained on large, diverse datasets (such as ImageNet with 1.28 million labelled images across 1,000 categories) are partially general-purpose: lower-layer features such as edge detectors, Gabor-like filters, and colour gradients transfer broadly across visual domains, while higher-layer features become increasingly task-specific and less transferable [14]. In the context of medical imaging, this observation motivates a two-phase training strategy: initialising a network with weights pretrained on natural images, then fine-tuning on the (smaller) medical dataset.

The degree to which natural image pretraining transfers to CT imaging depends on the architectural depth at which domain mismatch occurs. For shallow CNNs, the first few convolutional layers learn general low-level features (edges, textures) that are relevant to CT images; deeper layers trained for natural image classification may not transfer as effectively due to the fundamentally different statistical structure of CT images. Vision Transformers, however, exhibit a different transfer pattern: the self-attention mechanism learned on natural images captures relational patterns between patches that generalise more broadly than the rigid spatial inductive biases of convolution, potentially enabling better medical image representations from the same pretrained starting point.

Several studies have systematically investigated this transfer property for medical imaging. Pretraining on ImageNet consistently outperforms random initialisation as a starting point for medical image classification, detection, and segmentation tasks, even when the target domain (CT, MRI, histopathology) appears visually dissimilar to natural photographs. The benefit is especially pronounced for small medical datasets where training from random initialisation would overfit severely. In the CT IQA context, this justifies the use of ViT-B/32 pretrained on

ImageNet as the encoder backbone for CT-MUSIQ, despite the obvious domain gap between natural RGB photographs and single-channel greyscale CT images calibrated in Hounsfield Units.

### 2.10.2 Grayscale-to-RGB Adaptation Strategies

A practical challenge in transferring ImageNet-pretrained models to single-channel CT images is the three-channel (RGB) assumption embedded in the pretrained weights. Three principal strategies exist for handling this mismatch:

**Strategy 1: Channel repetition.** The greyscale image is replicated to all three channels, producing an RGB image where R = G = B = pixel value. This is the simplest approach and preserves the pretrained weights intact. The trade-off is that the model processes redundant information in all three channels, potentially wasting capacity on the trivial inter-channel correlations. However, for the patch embedding layer, channel repetition ensures that the linear combination of the three-channel input through the pretrained convolutional filter weights produces a meaningful activation, since the filter was designed to respond to spatially correlated patterns across channels.

**Strategy 2: Channel average initialisation.** The pretrained RGB patch embedding weights are collapsed to a single channel by averaging across the input channel dimension, and the single-channel architecture is used. This reduces the parameter count of the patch embedding layer by a factor of three but requires re-initialisation of the patch embedding, losing the pretrained weights.

**Strategy 3: Domain-specific pretraining.** The entire model is pretrained from scratch on large unlabelled medical image datasets using self-supervised objectives (e.g., masked image modelling as in MAE, or contrastive learning as in MoCo) before fine-tuning on the annotated CT quality labels. This approach is computationally expensive but potentially produces better initialised features for the CT domain.

CT-MUSIQ adopts Strategy 1 (channel repetition) as the most practical approach given the small 700-image training set and limited compute budget. The simplicity of this approach is supported by the empirical observation that even with channel repetition, the model achieves competitive quality assessment performance, suggesting that the pretrained ViT-B/32 features provide strong initialisation regardless of the greyscale-to-RGB conversion approach.

### 2.10.3 Fine-Tuning Stability on Small Medical Datasets

Fine-tuning large pretrained models on small datasets presents significant stability challenges. The key risks are: (1) catastrophic forgetting of pretrained representations, where aggressive fine-tuning overwrites the useful pretrained features; (2) overfitting to the limited training distribution; and (3) gradient instability due to the mismatch between the pretrained feature

scale and the randomly initialised task head gradients.

Several strategies have emerged to address these challenges in the medical imaging literature. Layer-wise learning rate decay, where earlier layers are fine-tuned with smaller learning rates than later layers, has been shown to preserve lower-level pretrained features while allowing higher-level task-specific adaptation. Prompt-based tuning approaches, which freeze the entire pretrained backbone and only train a small set of learnable prompt tokens injected into the input sequence, provide an extreme form of parameter-efficient adaptation that has shown promise for few-shot medical image analysis.

CT-MUSIQ addresses fine-tuning stability through the two-stage training protocol described in Chapter 3: Stage 1 warms up the randomly initialised domain-specific components (prediction heads, hash positional encoding) with the encoder frozen, preventing their noisy initial gradients from corrupting the pretrained attention patterns. Stage 2 then unfolds gradually with linear warm-up followed by cosine annealing, allowing the encoder to adapt smoothly to the CT domain while the heads continue to refine their quality mapping.

## 2.11 Regularisation Strategies for Small Dataset Deep Learning

The challenge of training deep learning models on small datasets like the 700-image CT-MUSIQ training set is a recurring theme in medical image analysis. This section reviews the principal regularisation strategies relevant to this setting, providing the motivation for the specific choices made in CT-MUSIQ.

### 2.11.1 Data Augmentation for CT Images

Data augmentation is the most widely used regularisation strategy for small medical datasets, as it artificially increases the effective training set size by applying label-preserving transformations to the original images. However, CT images impose significantly stricter constraints on acceptable augmentations than natural photographs, as discussed in Section 3.6.

Beyond the mild augmentations used in CT-MUSIQ (flips, small rotations, brightness jitter, Gaussian noise), more aggressive augmentation strategies such as CutMix, MixUp, and random erasing have been explored for medical image analysis. MixUp creates convex combinations of image-label pairs, which has been shown to improve calibration and reduce overconfident predictions. However, for CT IQA, the quality scores of mixed images do not follow a simple convex combination of the component scores (the perceptual quality of a noise-blended CT image is not predictable from the component scores), making MixUp semantically invalid without careful adaptation. The LDCTIQAC 2023 challenge top submission (agaldran) employed MixUp with specific adaptations, as reported in Galdran et al. [41], demonstrating that well-calibrated augmentation can improve performance on this dataset.

### 2.11.2 Dropout and Weight Decay

Dropout, originally introduced by Srivastava et al., is a widely used regularisation technique that randomly zeroes a fraction of activations during training, preventing co-adaptation of neurons and forcing more distributed feature representations. For Vision Transformers fine-tuned on small datasets, reducing the dropout rate below the standard 0.1 has been found to improve performance: the pretrained features already provide a form of implicit regularisation, and excessive dropout can prevent the model from fully leveraging the informative pretrained attention patterns.

Weight decay (L2 regularisation) applied through the decoupled AdamW optimiser provides an additional regularisation force that pushes parameter magnitudes toward zero, preventing overly large weights that generalise poorly. In CT-MUSIQ, weight decay is set to 0.01, a value that provides meaningful regularisation without preventing convergence.

### 2.11.3 Scale-Consistency KL Regularisation as a Novel Form of Self-Supervision

The scale-consistency KL divergence loss introduced in CT-MUSIQ can be understood as a form of self-supervised consistency regularisation, conceptually related to the consistency training paradigm in semi-supervised learning. In consistency training, the model is trained to produce similar predictions for two different augmented versions of the same input; this forces the model to learn representations that are invariant to the specific augmentation applied.

In CT-MUSIQ, the "two views" of the same CT image are the coarse-scale patches ($224 \times 224$) and the fine-scale patches ($384 \times 384$). These two views present the same underlying image content at different spatial resolutions, and the scale-consistency KL loss enforces that the model's quality assessment is consistent across these two views. This is semantically valid because the true diagnostic quality of a CT image does not depend on the resolution at which it is evaluated—a streak artefact visible at fine resolution is also present in the coarse-scale representation, merely at lower spatial frequency.

From a statistical learning theory perspective, the KL regularisation can be interpreted as an inductive bias toward scale-invariant quality representations. By penalising per-scale predictions that differ substantially from the global prediction, the loss encourages the model to learn features that are simultaneously visible and discriminative at both resolutions. This is particularly valuable for CT IQA because different classes of quality degradations manifest at different scales: global contrast reduction is visible at both scales equally, streak artefacts are more conspicuous at fine scale, while overall structural noise is better captured at coarse scale.

## 2.12 Evaluation Metrics in IQA Research

A comprehensive understanding of the evaluation metrics used in IQA research is essential for interpreting the experimental results in this thesis. This section reviews the theoretical properties of the three correlation metrics used in the LDCTIQAC 2023 challenge evaluation, with emphasis on what each metric measures and why their joint assessment provides a more complete picture of quality prediction performance.

### 2.12.1 Pearson Linear Correlation Coefficient

The Pearson Linear Correlation Coefficient (PLCC) measures the strength of the linear relationship between predicted and true quality scores. A PLCC of 1.0 indicates perfect linear agreement; PLCC of 0 indicates no linear relationship; PLCC of -1 indicates perfect negative linear correlation. The PLCC is sensitive to the scale and offset of the predictions: a model that systematically overestimates scores by a constant will have a high PLCC (if the linear relationship is preserved) but poor absolute accuracy.

In practice, PLCC rewards models that produce scores whose magnitude tracks the true quality scale accurately. It is the most natural metric for evaluating the quality of a regression model in the absolute sense, and is sensitive to the specific calibration of the prediction scale. A low PLCC may indicate either that the model cannot distinguish quality levels (poor rank ordering) or that its scores are poorly calibrated on the absolute scale (even if the relative ordering is correct).

### 2.12.2 Spearman Rank-Order Correlation Coefficient

The Spearman Rank-Order Correlation Coefficient (SROCC) measures the agreement of the rank orderings of predicted and true scores, independently of any linear or monotone rescaling of the predictions. Formally, SROCC is the Pearson correlation of the rank variables $R(\hat{q})$ and $R(q)$, where $R(\cdot)$ denotes the rank function.

SROCC is invariant to any monotone transformation of the predicted scores, making it a robust metric that does not penalise systematic scale errors. This property makes SROCC particularly appropriate as an evaluation metric for IQA, where the goal is to correctly order images by quality rather than to produce scores on a specific absolute scale. High SROCC indicates that the model can reliably order images from worst to best quality, which is the core requirement for practical applications such as scan rejection (reject scans below a threshold) and comparative benchmarking.

### 2.12.3 Kendall Rank-Order Correlation Coefficient

The Kendall Rank-Order Correlation Coefficient (KROCC) measures the proportion of concordant pairs (pairs where the model's ordering agrees with the true ordering) minus the

proportion of discordant pairs, normalised by the total number of pairs. It is a more granular measure of local ordering accuracy than SROCC: SROCC is determined by the global rank transformation, while KROCC counts each pair independently.

KROCC is generally harder to maximise than SROCC and is more sensitive to local ordering errors. A model with high SROCC but lower KROCC is one that gets the overall rank ordering roughly right but makes more errors at the pair level—for instance, it correctly identifies that group A images are higher quality than group B images, but makes more errors in distinguishing within groups. For clinical applications where the precise relative ordering of specific pairs of images matters (e.g., selecting the best scan from a repeated acquisition), KROCC is the most directly relevant metric.

### 2.12.4 Aggregate Score

The LDCTIQAC 2023 challenge uses the sum of absolute values of all three metrics (PLCC + SROCC + KROCC) as the primary ranking criterion, with a theoretical maximum of 3.0. This composite metric rewards models that perform well across all three aspects of quality prediction: absolute calibration (PLCC), global rank ordering (SROCC), and local pair ordering (KROCC). A model that excels in only one or two of these aspects will have a lower aggregate than one that performs well across all three, making the aggregate a comprehensive measure of overall IQA performance.

The aggregate metric also provides a natural weighting: since all three metrics have similar scales (all bounded in $[-1, 1]$), they contribute approximately equally to the aggregate, preventing any single metric from dominating the evaluation. This is particularly important when comparing methods with different architectures or training objectives, some of which may be more naturally suited to maximising one metric over the others.

# 3 Methodology

## 3.1 Overview and Problem Formulation

The central objective of this thesis is to develop an automated, no-reference image quality assessment (NR-IQA) system for brain low-dose computed tomography (LDCT) images. In clinical practice, radiologists manually evaluate each CT scan for diagnostic acceptability, a task that is time-consuming, subject to inter-rater variability, and unsuitable for high-throughput quality control pipelines. The proposed system, named CT-MUSIQ, aims to reproduce this human perceptual judgement automatically by learning a function that maps an input CT image directly to a continuous quality score, without requiring a paired reference scan of known perfect quality.

Formally, the task is defined as follows. Given a brain CT slice $\mathbf{x} \in \mathbb{R}^{H \times W}$ (where $H = W = 512$), the model must predict a scalar quality score $\hat{q} \in [0,4]$ that closely matches the consensus radiologist score $q \in [0,4]$ assigned on a Likert scale. The model has no access to the original, noise-free reference image during inference, making this a strictly no-reference problem. This is clinically important because in routine CT acquisition there is no reference available—the task is to assess perceptual quality from the single acquired image alone.

CT-MUSIQ is an architectural adaptation of MUSIQ (Multi-Scale Image Quality Transformer), a model proposed by Ke et al. [9] (ICCV 2021) for natural image quality assessment. The key innovations introduced in CT-MUSIQ over the vanilla MUSIQ architecture are: (1) a custom hash-based positional encoding scheme tailored to multi-scale variable-length patch sequences, (2) per-scale prediction heads that enable a novel scale-consistency KL divergence regularisation loss, (3) a pairwise margin ranking loss for improved ordinal discrimination, and (4) a clinically-motivated brain CT windowing pre-processing step in place of standard natural image normalisation. The complete system is trained end-to-end on the LDCTIQAC 2023 challenge dataset [30] using a two-stage fine-tuning protocol with mixed-precision training, designed to fit within 6 GB of GPU video memory (VRAM) on a consumer-grade NVIDIA RTX 3060 Laptop GPU.

The following sections describe each methodological component in detail: the dataset and pre-processing pipeline (Section 3.2), the CT-MUSIQ model architecture (Section 3.3), the composite loss function (Section 3.4), the training procedure (Section 3.5), data augmentation (Section 3.6), the evaluation protocol (Section 3.7), the comparison baseline (Section 3.8), and the ablation study design (Section 3.9).

## 3.2 Dataset and Pre-Processing

### 3.2.1 LDCTIQAC 2023 Dataset

All experiments in this thesis are conducted on the **LDCTIQAC 2023 dataset** (Low-Dose CT Image Quality Assessment Challenge 2023), which was the basis of a challenge organised at MICCAI 2023 and subsequently documented in Lee et al. [30] in *Medical Image Analysis*. The dataset consists of **1,000 axial brain CT slices**, each stored as a 32-bit floating-point TIFF image of dimensions $512 \times 512$ pixels. The pixel values in the distributed files are pre-normalised to the range $[0, 1]$ prior to release. Each image corresponds to a single axial slice from a low-dose CT acquisition protocol designed to simulate clinically realistic noise levels encountered in routine brain CT examinations.

Quality labels were obtained through expert radiologist assessment. Multiple radiologists independently assigned a perceptual quality score to each image using a five-point Likert scale ranging from 0 (completely unacceptable, severe noise or artefacts) to 4 (excellent quality, fully diagnostic). The labels distributed with the dataset represent the averaged consensus score across all annotating radiologists, yielding a continuous floating-point value for each image. This averaged score is used as the regression target throughout training and evaluation.

The image files follow a strict four-digit zero-padded naming convention: `{index:04d}.tif`, yielding filenames `0000.tif` through `0999.tif`. The mapping from filenames to quality scores is provided in a JSON file (`train.json`) with the filename (including extension) as the key and the quality score as the value.

### 3.2.2 Data Splits

The 1,000 images are partitioned into three non-overlapping subsets based on filename index order, as specified in the challenge protocol: a **training set** comprising images 0000 through 0699 (700 images, 70%), a **validation set** comprising images 0700 through 0899 (200 images, 20%), and a **test set** comprising images 0900 through 0999 (100 images, 10%).

Importantly, the split is deterministic and index-based rather than randomly sampled. This design choice ensures that all reported results are reproducible without knowledge of a random seed for splitting, and it matches the official challenge protocol. During training, only the training split is used for gradient updates. The validation split is used exclusively for hyperparameter tuning, model selection (checkpoint saving based on best validation aggregate score), and early stopping. The test split is held out completely until final evaluation and is never used to inform any training decision.

### 3.2.3 CT Windowing Pre-Processing

CT images store raw pixel values in Hounsfield Units (HU), a quantitative scale where water is 0 HU, air is approximately $-1000$ HU, and dense bone is approximately $+1000$ HU. Raw HU values span a large dynamic range (typically $-1024$ to $+3071$ HU) that is clinically irrelevant for any single tissue type. Radiologists routinely apply a **CT window** transformation before viewing: they select a window level (centre) and window width that clips and linearly rescales the HU range to display only the tissue of interest with full contrast.

A critical methodological adaptation in this thesis concerns the windowing parameters. The original LDCTIQAC 2023 challenge was designed around **abdominal CT** imaging, for which the clinical standard window is centred at level 40 HU with a width of 350 HU. However, the specific dataset used in this thesis contains **brain CT slices**, for which the standard clinical window is the **brain soft-tissue window**: level 40 HU, width 80 HU. This choice reflects the practice of neuroradiology: the narrow brain window (spanning from 0 to 80 HU) optimally differentiates grey matter ($\sim$40 HU), white matter ($\sim$35 HU), cerebrospinal fluid ($\sim$0–15 HU), and hyperacute haemorrhage ($\sim$50–80 HU), which are the diagnostically relevant tissues in brain CT quality assessment.

The window transformation is applied as follows. Given an input pixel value $p$ and window parameters level $L = 40$ HU and width $W = 80$ HU:

$$\mathrm{HU}_{\min} = L - \frac{W}{2} = 40 - 40 = 0 \text{ HU} \tag{3.1}$$

$$\mathrm{HU}_{\max} = L + \frac{W}{2} = 40 + 40 = 80 \text{ HU} \tag{3.2}$$

$$p_{\text{clipped}} = \text{clip}(p,\ \mathrm{HU}_{\min},\ \mathrm{HU}_{\max}) \tag{3.3}$$

$$p_{\text{norm}} = \frac{p_{\text{clipped}} - \mathrm{HU}_{\min}}{\mathrm{HU}_{\max} - \mathrm{HU}_{\min}} \in [0,1] \tag{3.4}$$

Since the dataset images are already stored as pre-normalised float32 values in $[0,1]$, the windowing is implicitly applied during dataset generation. No further normalisation using ImageNet mean and standard deviation is applied, as the images are CT scans rather than natural images.

### 3.2.4 Data Loading Pipeline

The `LDCTDataset` PyTorch `Dataset` class implements the complete image loading and pre-processing pipeline for a given split. During a single dataset access, the following operations are performed in sequence:

1. **Image loading**: The 32-bit float TIFF file is opened with PIL, converted to a NumPy

array of shape $(512, 512)$, and clipped to $[0, 1]$.

2. **Augmentation** (training split only): stochastic spatial and photometric transformations are applied (see Section 3.6).
3. **Multi-scale pyramid construction**: The image is resized to each target scale using bicubic interpolation.
4. **Patch extraction**: Each resized image is divided into non-overlapping $32 \times 32$ patches, and the spatial coordinates (scale index, row index, column index) of each patch are recorded.
5. **Channel replication**: Each greyscale patch is replicated into three identical channels to produce an RGB-compatible tensor.
6. **Tensor conversion**: All patches and coordinates are converted to PyTorch tensors.

The `DataLoader` is configured with `batch_size = 10`, `shuffle = True` for training and `shuffle = False` for validation and test, `num_workers = 0` for Windows-compatible multiprocessing, `pin_memory = True` for efficient GPU transfer, and `drop_last = True` for training.

## 3.3 Proposed Model Architecture: CT-MUSIQ

### 3.3.1 Architecture Overview

CT-MUSIQ is a Transformer-based no-reference image quality assessor that operates on multi-scale non-overlapping image patches. Inspired by the MUSIQ model of Ke et al. [9], which demonstrated that a Vision Transformer processing patches from images at multiple resolutions simultaneously can learn richer quality-relevant representations than single-scale CNNs, CT-MUSIQ adapts this paradigm for the specific challenges of brain CT quality assessment: single-channel greyscale images, small dataset size (700 training samples), and tight GPU memory constraints.

The architecture consists of five principal components, executed in sequence during each forward pass:

1. **Multi-scale patch tokenisation**: The input CT image is resized to two scales ($224 \times 224$ and $384 \times 384$) and divided into non-overlapping $32 \times 32$ patches, yielding 49 and 144 patches respectively, for a total of 193 patch tokens per image.
2. **Patch Embedding**: Each patch is projected from pixel space ($3 \times 32 \times 32$) to a 768-dimensional embedding space using a shared 2D convolutional projection layer initialised with pretrained ViT-B/32 weights.
3. **Hash-based Positional Encoding**: Spatial position and scale identity are injected into each token via a sum of three learned embedding lookups.
4. **Transformer Encoder**: A 12-layer, 8-head Transformer encoder processes the full

sequence of 193 patch tokens prepended with a learnable `[CLS]` token (194 tokens total), performing global self-attention.

5. **Prediction Heads**: A two-layer MLP global head reads the `[CLS]` token to produce the final quality score. Two auxiliary per-scale heads read scale-averaged token representations to produce scale-specific scores used only during training for the KL consistency loss.

### 3.3.2 Multi-Scale Patch Tokenisation

The motivation for multi-scale representation in IQA is well established: quality degradations such as noise, blur, and compression artefacts manifest differently at different spatial frequencies and resolutions. A patch at a coarse scale captures global structural context, while a patch at a fine scale captures local texture and noise statistics. Processing both simultaneously enables the model to integrate quality cues across spatial scales.

Two target scales are used: $S_1 = 224$ pixels and $S_2 = 384$ pixels. Each CT image ($512 \times 512$ pixels) is independently resized to each target scale using bicubic interpolation. Each resized image is then divided into a regular grid of non-overlapping $32 \times 32$ pixel patches:

- At scale 224: grid size $= 224 \div 32 = 7$, yielding $7 \times 7 = 49$ patches.
- At scale 384: grid size $= 384 \div 32 = 12$, yielding $12 \times 12 = 144$ patches.
- Total patch count: $49 + 144 = 193$ patches per image.

The choice of patch size $P = 32$ is motivated by compatibility with the pretrained Vision Transformer ViT-B/32, which uses $32 \times 32$ patches. Patches from both scales are concatenated into a single sequence (scale-0 patches first, then scale-1 patches).

### 3.3.3 Spatial Coordinate Representation

Each patch is assigned a three-dimensional coordinate vector $(s_i, r_i, c_i)$ where $s_i \in \{0,1\}$ is the scale index, $r_i \in \{0,\ldots,g_s - 1\}$ is the row index within the grid of scale $s$, and $c_i \in \{0,\ldots,g_s - 1\}$ is the column index, with $g_s = S_s \div P$ being the grid size at scale $s$. For the 193 total patches, the coordinate array has shape $[193, 3]$. These coordinates are stored as integer tensors and consumed by the hash-based positional encoding module.

### 3.3.4 Grayscale-to-RGB Channel Replication

The pretrained ViT-B/32 model was trained on ImageNet RGB images, so its patch embedding convolution has shape $[768, 3, 32, 32]$. Brain CT images are inherently single-channel (greyscale). To exploit the pretrained patch embedding weights without modification, each greyscale patch of shape $[1, 32, 32]$ is replicated into three identical channels to produce a tensor of shape $[3, 32, 32]$:

$$\mathbf{p}_{\text{RGB}} = [p, p, p]_{\text{channels}} \tag{3.5}$$

This replication means the embedding layer can be initialised with meaningful pretrained features rather than requiring random initialisation, which is especially valuable given the small training dataset of 700 images. No artificial colour information is introduced.

### 3.3.5 Patch Embedding Module

The patch embedding module is a single 2D convolutional layer with kernel size equal to the patch size ($P = 32$) and stride equal to $P$. For a batch of patches of shape $[B \cdot N, 3, 32, 32]$, the convolution produces $[B \cdot N, 768, 1, 1]$. After flattening, each patch is represented as a 768-dimensional embedding vector. Formally, for patch tensor $\mathbf{p} \in \mathbb{R}^{3\times32\times32}$:

$$\mathbf{e} = \text{Flatten}\big(\text{Conv2d}(\mathbf{p})\big) \in \mathbb{R}^{768} \tag{3.6}$$

The projection weights are initialised with pretrained ViT-B/32 weights from the *timm* library (`vit_base_patch32_224`).

### 3.3.6 Hash-Based Positional Encoding

Standard sinusoidal or learned positional encodings in Vision Transformers assume a fixed sequence length. CT-MUSIQ processes variable-composition sequences of 193 tokens (49 from scale 0 plus 144 from scale 1), and the distinction between scales is crucial for the model to exploit scale-specific features.

The custom hash positional encoding module addresses this by decomposing position into three orthogonal components, each represented by an independent learnable embedding table:

$$\mathbf{pos}_i = \mathbf{E}_{\text{scale}}(s_i) + \mathbf{E}_{\text{row}}(r_i) + \mathbf{E}_{\text{col}}(c_i) \tag{3.7}$$

where $\mathbf{E}_{\text{scale}} \in \mathbb{R}^{n_s\times d}$, $\mathbf{E}_{\text{row}} \in \mathbb{R}^{G_{\max}\times d}$, and $\mathbf{E}_{\text{col}} \in \mathbb{R}^{G_{\max}\times d}$ are three learned embedding matrices; $n_s = 2$ is the number of scales; $G_{\max} = 13$ is the maximum grid dimension (accommodating up to $13 \times 13$ patches, covering the $384/32 = 12$ grid size with margin); and $d = 768$ is the embedding dimension. All embedding tables are initialised with zero-mean Gaussian noise with standard deviation 0.02.

This decomposition has several desirable properties. First, it handles the variable-length multi-scale sequence without requiring padding. Second, the scale embedding $\mathbf{E}_{\text{scale}}(s)$ allows the model to learn scale-specific biases. Third, the factored structure dramatically reduces the embedding table size to $(2+13+13) \times 768 = 21{,}504$ parameters versus $2 \times 13 \times 13 \times 768 = 328{,}704$ for a full 3D lookup table. The positional encoding is added element-wise to the

patch embeddings:

$$\mathbf{z}_i = \mathbf{e}_i + \mathbf{pos}_i, \quad i = 1, \ldots, N \tag{3.8}$$

### 3.3.7 Learnable [CLS] Token

Following the standard Vision Transformer design, a learnable global classification token ([CLS] token) is prepended to the patch token sequence. The [CLS] token is a single parameter vector $\theta_{\text{cls}} \in \mathbb{R}^{768}$, initialised from $\mathcal{N}(0, 0.02^2)$ and replicated across the batch dimension during the forward pass. After the Transformer encoder processes the full sequence of $N+1 = 194$ tokens, the [CLS] token at position 0 in the output sequence aggregates global context from all 193 patch tokens via the self-attention mechanism.

### 3.3.8 Transformer Encoder

The Transformer encoder is a stack of $L = 12$ encoder layers. Each layer operates on the full sequence of 194 tokens and performs the following operations:

1. **Pre-Layer Normalisation**: Layer normalisation is applied before each sub-layer ("norm-first" configuration), which improves training stability when fine-tuning on small datasets.
2. **Multi-Head Self-Attention**: With $H = 8$ attention heads and $d_{\text{model}} = 768$, each head has per-head dimension $d_k = 768/8 = 96$. Global self-attention is performed over all 194 tokens, enabling the model to capture long-range spatial dependencies across both scales simultaneously.
3. **Feed-Forward Network (FFN)**: A position-wise FFN with hidden dimension $d_{\text{ff}} = 3072 = 4 \times d_{\text{model}}$, using GELU activation and dropout $p = 0.05$.
4. **Residual Connections**: Standard residual connections around each sub-layer.

The dropout rate is deliberately reduced to $p = 0.05$ (versus the standard 0.1 in ViT) to mitigate overfitting on the small 700-sample training set. The encoder weights are initialised from pretrained ViT-B/32 weights, covering all attention projection weights, layer normalisation parameters, and FFN weights for all 12 layers. The total number of Transformer encoder parameters is approximately 86 million (matching ViT-B/32).

### 3.3.9 Prediction Heads

**Global Head:** The [CLS] token output $\mathbf{c} \in \mathbb{R}^{768}$ is passed through a two-layer MLP:

$$\hat{q} = W_2 \cdot \text{Dropout}\big(\text{GELU}(W_1 \mathbf{c})\big) \tag{3.9}$$

where $W_1 \in \mathbb{R}^{384\times768}$ and $W_2 \in \mathbb{R}^{1\times384}$. The output is a single scalar $\hat{q} \in \mathbb{R}$, which is the predicted quality score at inference time.

**Per-Scale Heads:** For each scale $s \in \{0,1\}$, the patch token outputs corresponding to that scale are extracted and mean-pooled along the patch dimension to produce a scale-specific representation $\mathbf{c}_s \in \mathbb{R}^{768}$. This is passed through an identical two-layer MLP to produce a per-scale quality score $\hat{q}_s$. The per-scale scores are used exclusively during training to compute the scale-consistency KL divergence loss (Section 3.4) and are discarded at inference time.

**Parameter count summary:**

- Patch embedding: $768 \times 3 \times 32 \times 32 = 2{,}359{,}296$ params
- Hash positional encoding: $28 \times 768 = 21{,}504$ params
- CLS token: 768 params
- Transformer encoder (12 layers): $\approx 85{,}054{,}464$ params
- Global head: $384 \times 768 + 1 \times 384 = 295{,}296$ params
- Per-scale heads (2): $2 \times 295{,}296 = 590{,}592$ params
- **Total:** $\approx 88.3$ **million parameters**

## 3.4 Loss Function

CT-MUSIQ is trained with a composite loss function combining three terms: a primary regression loss, a scale-consistency regularisation loss, and a pairwise ranking loss. The combined objective is:

$$\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{MSE}} + \lambda_{\text{KL}} \cdot \mathscr{L}_{\text{KL}} + \lambda_{\text{rank}} \cdot \mathscr{L}_{\text{rank}} \tag{3.10}$$

where $\lambda_{\text{KL}} = 0.10$ (determined by ablation study, Section 3.9) and $\lambda_{\text{rank}} = 0.5$.

### 3.4.1 Mean Squared Error Loss

The primary regression objective is the standard mean squared error (MSE) between the global predicted score $\hat{q}$ and the ground truth radiologist score $q$:

$$\mathscr{L}_{\text{MSE}} = \frac{1}{B}\sum_{i=1}^{B}\left(\hat{q}_i - q_i\right)^2 \tag{3.11}$$

MSE penalises large errors quadratically, encouraging the model to be accurate for images whose quality is clearly very high or very low, which is clinically most important. While MSE is a standard regression objective, it does not explicitly penalise incorrect pairwise orderings, motivating the addition of the ranking loss component.

### 3.4.2 Scale-Consistency KL Divergence Loss

The key regularisation innovation in CT-MUSIQ is the **scale-consistency KL divergence loss**, which penalises disagreement between the per-scale quality predictions ($\hat{q}_s$) and the global quality prediction ($\hat{q}$). The intuition is that if both scales observe the same image at different resolutions, their quality assessments should be broadly consistent. A model that learns scale-specific shortcuts is learning artefacts rather than true quality features—a particular risk given the small training set of only 700 images.

Computing KL divergence between scalar scores requires converting them to probability distributions. This is accomplished via a **Gaussian kernel smoothing** procedure over $K = 20$ discrete bins uniformly spanning the score range $[0,4]$. For a scalar score $\hat{q}$ and bin centres $\{b_k\}_{k=1}^{K}$:

$$\tilde{p}_k(\hat{q}) = \exp\left(-\frac{(\hat{q}-b_k)^2}{2\sigma^2}\right) \tag{3.12}$$

$$p_k(\hat{q}) = \frac{\tilde{p}_k(\hat{q})}{\sum_{j=1}^{K} \tilde{p}_j(\hat{q}) + \varepsilon} \tag{3.13}$$

with $\sigma = 0.5$ and $\varepsilon = 10^{-8}$ for numerical stability. The result is a soft probability distribution $\mathbf{p}(\hat{q}) \in \Delta^{K-1}$ that peaks at the bin nearest to $\hat{q}$ and decays smoothly away. The scale-consistency loss is then:

$$\mathscr{L}_{\mathrm{KL}} = \frac{1}{n_s} \sum_{s=0}^{n_s-1} D_{\mathrm{KL}}\Big(\mathbf{p}(\hat{q}_s) \parallel \mathbf{p}(\hat{q})\Big) \tag{3.14}$$

where $D_{\mathrm{KL}}$ denotes the KL divergence computed using log-probabilities of the scale distribution against the probability of the global distribution, with batch-mean reduction. The global prediction is treated as the target distribution; the per-scale predictions are the "approximating" distributions. The asymmetry of KL divergence means that this loss severely penalises per-scale predictions that assign high probability to bins that the global head considers very unlikely, preventing extreme scale-specific outlier predictions.

### 3.4.3 Pairwise Margin Ranking Loss

The MSE loss optimises absolute score accuracy but does not explicitly penalise incorrect pairwise orderings. Since the LDCTIQAC 2023 evaluation metrics (PLCC, SROCC, KROCC) all measure rank-based correlation, training with an explicit ranking objective is expected to improve performance. The **pairwise margin ranking loss** operates on all $B(B-1)$ ordered

pairs $(i, j)$ in each batch of size $B$:

$$\mathscr{L}_{\text{rank}} = \frac{1}{B^2} \sum_{i=1}^{B} \sum_{j=1}^{B} \max\Big(0,\ m - \text{sgn}(q_i - q_j) \cdot (\hat{q}_i - \hat{q}_j)\Big) \tag{3.15}$$

where $m = 0.1$ is the margin. The term $\text{sgn}(q_i - q_j)$ is $+1$ if image $i$ truly has higher quality than image $j$, $-1$ if lower, and 0 if equal. The loss encourages the predicted score difference $\hat{q}_i - \hat{q}_j$ to have the same sign as the true difference and to exceed the margin $m$, directly optimising relative ordering that translates to improved rank correlation metrics.

The batch-level pairwise formulation is computationally efficient: for a batch of $B = 10$ images, there are $10 \times 10 = 100$ pairs, and the outer product over predicted and target differences can be computed as matrix operations. The diagonal terms (where $i = j$) contribute zero loss because $\text{sgn}(0) = 0$.

## 3.5 Training Procedure

### 3.5.1 Two-Stage Training Schedule

CT-MUSIQ employs a two-stage training strategy designed to stabilise fine-tuning of pre-trained ViT weights on a small medical imaging dataset.

**Stage 1 (Epochs 1–5): Frozen Encoder Warm-Up.** In the first stage, the Transformer encoder weights and the `[CLS]` token parameter are frozen. Only the patch embedding layer, hash positional encoding, and prediction heads are trained. The learning rate is set to $\text{LR}_{\text{stage1}} = 3 \times 10^{-4}$, which is relatively high to allow the randomly-initialised CT-specific components to reach a useful parameter regime quickly. The rationale is to prevent the noisy gradients from randomly-initialised prediction heads from corrupting the carefully pretrained attention patterns during the earliest training steps.

**Stage 2 (Epochs 6–100): Full Fine-Tuning.** Beginning at epoch 6, all parameters are unfrozen. The learning rate transitions through two sub-phases within Stage 2:

1. **Linear Warm-Up (Epochs 6–8)**: The learning rate is linearly increased from $\text{LR}_{\text{stage2}} \cdot (1/3)$ to $\text{LR}_{\text{stage2}} = 5 \times 10^{-5}$ over 3 epochs. This warm-up prevents the abrupt large-gradient update that would occur if the encoder were suddenly fully unfrozen at full learning rate.
2. **Cosine Annealing (Epochs 9–100)**: After the warm-up, a cosine annealing scheduler gradually decays the learning rate from $5 \times 10^{-5}$ to $\text{LR}_{\text{min}} = 10^{-6}$:

$$\text{LR}(t) = \text{LR}_{\text{min}} + \frac{1}{2}(\text{LR}_{\text{max}} - \text{LR}_{\text{min}}) \left(1 + \cos\left(\frac{\pi t}{T_{\text{max}}}\right)\right) \tag{3.16}$$

where $t$ is the number of epochs elapsed since the start of the cosine phase and $T_{\max}$ is the total number of cosine epochs.

### 3.5.2 Optimiser

The AdamW optimiser (Adam with decoupled weight decay) is used throughout training with weight decay coefficient $\lambda_{\text{wd}} = 0.01$. AdamW is preferred over standard Adam for fine-tuning pretrained Transformers because decoupled weight decay provides more principled regularisation: it applies decay uniformly to all parameters rather than scaling it by the adaptive learning rate per parameter, which can distort the effective regularisation strength when learning rates vary across parameters due to gradient magnitude adaptation.

### 3.5.3 Mixed Precision Training

To fit the 12-layer ViT-B-based model with batch size 10 within 6 GB of GPU VRAM, **automatic mixed precision (AMP)** training is employed using PyTorch's `torch.cuda.amp` framework. AMP operates such that the forward pass and loss computation use half-precision (fp16) tensors for activations, while the backward pass and parameter updates are performed in full precision (fp32). The approximate VRAM breakdown with AMP is: model weights $\approx$ 1.5 GB, activations $\approx$ 1.5–2 GB, and optimiser states $\approx$ 1.5 GB, for a total well within the 6 GB budget. Mixed precision saves approximately 40% of VRAM relative to pure fp32 and provides approximately 30% speedup on the RTX 3060 GPU.

A `GradScaler` is used to dynamically scale the loss before the backward pass to prevent fp16 gradient underflow. The scaler reduces the scale factor if overflow is detected and gradually increases it otherwise.

### 3.5.4 Gradient Clipping

Gradient clipping is applied after the `GradScaler.unscale_()` step, which restores gradients to fp32 scale before clipping. The maximum gradient norm is $\|\nabla\|_{\max} = 1.0$, applied to all trainable parameters jointly using the global $\ell_2$ norm. This prevents the occasional large gradient updates that can occur at the Stage 1 to Stage 2 transition when the previously frozen encoder is suddenly exposed to the full learning rate.

### 3.5.5 Exponential Moving Average (EMA)

An **Exponential Moving Average (EMA)** of the model weights is maintained during training with decay factor $\rho = 0.999$:

$$\theta_{\text{EMA}}^{(t)} = \rho \cdot \theta_{\text{EMA}}^{(t-1)} + (1-\rho) \cdot \theta^{(t)} \tag{3.17}$$

During validation at the end of each epoch, the EMA weights are temporarily substituted

into the model, the validation pass is performed, and then the live weights are restored. The checkpoint saved when validation performance improves stores the EMA weights. This ensures the saved model benefits from the smoothing effect of EMA, which tends to have better generalisation properties because it effectively averages out the stochastic trajectory of stochastic gradient descent.

### 3.5.6 Early Stopping

Training is monitored against the validation aggregate score $|\text{PLCC}| + |\text{SROCC}| + |\text{KROCC}|$. If the validation aggregate does not improve for `PATIENCE` $= 20$ consecutive epochs, training is halted early to prevent overfitting. In all full training runs conducted for this thesis, early stopping was not triggered before epoch 100, indicating that the model continued improving through the full training budget.

## 3.6 Data Augmentation

Given the small training set of 700 images, data augmentation is essential to improve generalisation and prevent overfitting. CT imaging places unique constraints on acceptable augmentations: unlike natural images, CT images are calibrated physical measurements in Hounsfield Units, and their noise texture and structural patterns are diagnostically meaningful. Aggressive colour distortions or large geometric deformations would alter the perceived quality in ways that do not correspond to any real imaging artefact, creating label noise.

Therefore, augmentations are carefully chosen to be mild and clinically plausible. The following augmentations are applied stochastically during training only (never during validation or testing):

- **Horizontal Flip** ($p = 0.5$): The CT image is flipped left-to-right. Axial brain CT slices are broadly symmetric across the mid-sagittal plane, making horizontal flip a valid augmentation that does not alter quality-relevant statistics.
- **Vertical Flip** ($p = 0.5$): The CT image is flipped top-to-bottom. Qualitative characteristics of noise and artefacts are isotropic in axial slices, making vertical flip reasonable.
- **Small Random Rotation** ($p = 0.5$, angle $\in [-7, +7]$): Applied using bilinear interpolation in float-mode PIL to avoid 8-bit quantisation artefacts. The small angular range preserves anatomical plausibility.
- **Random Crop and Resize** ($p = 0.5$, crop fraction $\in [0.85, 1.0]$): A random crop retaining 85–100% of the original image area is taken and bicubically resized back to $512 \times 512$. This simulates slight variation in the field of view without removing diagnostically critical anatomy.
- **Mild Brightness and Contrast Jitter** (delta $\in [-0.05, +0.05]$): Brightness is perturbed by a random additive offset and contrast by a multiplicative factor, both sampled from

$[-0.05, +0.05]$. This range is deliberately small to avoid shifting the effective HU mapping significantly.

- **Mild Additive Gaussian Noise** (std $\in [0.005, 0.02]$): Random Gaussian noise with standard deviation uniformly sampled from $[0.005, 0.02]$ is added. This simulates scanner noise variability between similar low-dose acquisitions.

All augmentations are applied in float32 precision, and the result is clipped to $[0, 1]$ after all transformations to maintain valid intensity range.

## 3.7 Evaluation Protocol

### 3.7.1 Evaluation Metrics

The evaluation metrics precisely follow the LDCTIQAC 2023 challenge protocol as described in Lee et al. [30]. Three correlation coefficients are computed between the predicted scores and the ground truth scores:

**Pearson Linear Correlation Coefficient (PLCC):**

$$\text{PLCC} = \frac{\sum_i (\hat{q}_i - \bar{\hat{q}})(q_i - \bar{q})}{\sqrt{\sum_i (\hat{q}_i - \bar{\hat{q}})^2 \sum_i (q_i - \bar{q})^2}} \tag{3.18}$$

**Spearman Rank-Order Correlation Coefficient (SROCC):**

$$\text{SROCC} = 1 - \frac{6 \sum_i d_i^2}{N_{\text{test}}(N_{\text{test}}^2 - 1)} \tag{3.19}$$

where $d_i$ is the difference in ranks between $\hat{q}_i$ and $q_i$.

**Kendall Rank-Order Correlation Coefficient (KROCC):**

$$\text{KROCC} = \frac{n_c - n_d}{\frac{1}{2} N_{\text{test}}(N_{\text{test}} - 1)} \tag{3.20}$$

where $n_c$ is the number of concordant pairs (pairs where the predicted and true orderings agree) and $n_d$ is the number of discordant pairs.

**Aggregate Score:**

$$\text{Aggregate} = |\text{PLCC}| + |\text{SROCC}| + |\text{KROCC}| \tag{3.21}$$

The aggregate score is the primary ranking metric used in the LDCTIQAC 2023 challenge leaderboard, with a maximum possible value of 3.0 (perfect correlation on all three metrics simultaneously).

**On metric complementarity:** PLCC measures linear correlation of the predicted scores with the ground truth and is sensitive to the absolute scale and offset of predictions. SROCC measures the agreement of rank orderings and is invariant to monotone transformations of the predicted scores. KROCC measures concordance at the pair level and is more sensitive to local ordering errors than SROCC. Using all three jointly as an aggregate provides a more comprehensive picture of model quality than any single metric alone.

### 3.7.2 Test-Time Augmentation

Test-time augmentation (TTA) with horizontal flip averaging is implemented as an optional inference strategy. For each test image, the model is run twice: once on the original patch sequence and once on a horizontally-flipped version. The horizontal flip remaps column coordinates as:

$$c_{\text{flipped}} = (g_s - 1) - c_{\text{original}} \tag{3.22}$$

where $g_s$ is the grid size at scale $s$. The final prediction is the average of the two forward pass outputs. TTA exploits the approximate left-right symmetry of brain CT images to reduce the variance of the quality score estimate.

### 3.7.3 Isotonic Regression Calibration

An optional post-processing calibration step using **isotonic regression** is implemented. Isotonic regression fits a non-decreasing step function mapping raw predictions to calibrated outputs, fitted on the validation set:

$$\hat{q}^*_{\text{test}} = \text{ISO}(\hat{q}_{\text{test}}) \tag{3.23}$$

where ISO is the isotonic regression function fitted on validation set pairs $\{(\hat{q}_{\text{val},i}, q_{\text{val},i})\}$. Because isotonic regression is a monotone transformation, it preserves rank order exactly (SROCC and KROCC are unchanged) while potentially improving PLCC by correcting systematic scale or bias errors in the raw output.

## 3.8 Comparison Baseline: Agaldran Combo

For a fair within-paper comparison, a strong baseline model is trained and evaluated under identical conditions. The baseline, termed **Agaldran Combo**, is motivated by the top-performing method in the LDCTIQAC 2023 challenge, which combined a Swin Transformer with a ResNet backbone [30, 41].

The baseline model is a **score-level ensemble** of two independently pretrained backbones:

1. **Swin Transformer Tiny (Swin-T)**: Loaded from *timm* (`swin_tiny_patch4_window7_224`) with ImageNet-1K pretrained weights. The classification head is removed and replaced with a quality prediction head: Linear(768, 256) → ReLU → Dropout(0.05) → Linear(256, 1).
2. **ResNet-50**: Loaded from *torchvision* with ImageNet-1K pretrained weights. The fully connected classification layer is replaced by a quality prediction head: Linear(2048, 256) → ReLU → Dropout(0.05) → Linear(256, 1).

The final prediction is the arithmetic mean of the two backbones' quality scores:

$$\hat{q}_{\text{combo}} = \frac{\hat{q}_{\text{Swin-T}} + \hat{q}_{\text{ResNet50}}}{2} \tag{3.24}$$

The baseline accepts full CT images as input (not patches) of shape $[B, 3, H, W]$. Each greyscale CT image is replicated to three channels using the same channel replication strategy as CT-MUSIQ.

The baseline is trained under the same loss function (`CTMUSIQLoss`), batch size (10), and dataset split as CT-MUSIQ. Unlike CT-MUSIQ, the baseline does not employ the two-stage training schedule; it is trained end-to-end from the start with a constant learning rate of $3 \times 10^{-4}$ and the same early stopping patience of 20 epochs. This design choice allows the baseline to be more straightforwardly trained while still being evaluated under identical conditions.

## 3.9 Ablation Study Design

To isolate the contribution of each proposed component, a systematic ablation study is designed with six configurations (A1–A6), each removing or modifying one aspect of the full CT-MUSIQ model:

**A1: Single-scale baseline (scale 224 only, no KL loss)** Tests the importance of multi-scale processing by restricting to only the coarse scale, with all other components identical to the full model.

**A2: Two scales, no KL loss** Tests the contribution of the KL scale-consistency loss by disabling it ($\lambda_{\text{KL}} = 0$) while keeping the full multi-scale input.

**A3: Two scales, KL $\lambda = 0.05$** Tests a weaker KL weight than the proposed $\lambda_{\text{KL}} = 0.10$.

**A4: Two scales, KL $\lambda = 0.10$** The proposed full configuration, included in the ablation table for direct comparison.

**A5: Two scales, KL $\lambda = 0.20$** Tests whether a stronger KL weight over- regularises the model.

**A6: Three scales** $(224, 384, 512)$**, KL $\lambda = 0.10$** Tests whether adding a third, full-resolution

scale improves performance. This configuration requires more VRAM and extends the positional encoding to $G_{\max} = 17$.

All ablation runs are trained for 30 epochs (a shortened budget to allow multiple configurations to be evaluated within the compute budget), using the same training procedure, data splits, and random seed as the full CT-MUSIQ model. The validation aggregate at epoch 30 is used as the comparison metric across ablation configurations.

# 4 Experiments

## 4.1 Overview

This chapter provides a comprehensive description of the experimental framework used to evaluate CT-MUSIQ and to validate the contributions of each proposed architectural and training component. Section 4.2 details the software and hardware environment. Section 4.3 documents the full set of hyperparameters used for both CT-MUSIQ and the Agaldran Combo baseline. Section 4.4 describes the main training runs in chronological order. Section 4.5 presents the complete ablation study protocol and results. Section 4.6 specifies the evaluation procedure for the held-out test set. Section 4.7 discusses reproducibility provisions.

## 4.2 Implementation Details

### 4.2.1 Software Environment

The complete CT-MUSIQ project is implemented in Python 3.12 running in a virtual environment (`.venv`) created with `py -3.12 -m venv .venv`. The following core dependencies are used:

- **PyTorch 2.x**: Primary deep learning framework for model definition, training loop, automatic differentiation, and CUDA acceleration.
- **timm ($\geq$ 0.9.0)**: Model zoo library providing pretrained ViT-B/32 weights (`vit_base_patch32_224`) and Swin-T weights (`swin_tiny_patch4_window7_224`).
- **torchvision ($\geq$ 0.15)**: Used for pretrained ResNet-50 weights in the Agaldran Combo baseline.
- **PIL/Pillow**: Image loading and float-mode augmentation operations, including bicubic resizing and rotation without quantisation artefacts.
- **NumPy**: Array operations during preprocessing, augmentation, and metric computation.
- **SciPy**: Pearson, Spearman, and Kendall correlation coefficient computation for evaluation.
- **scikit-learn**: Isotonic regression calibration for the optional post-processing step.
- **pandas**: Training log CSV management for epoch-by-epoch logging.
- **tqdm**: Progress bar display during training and evaluation.

All hyperparameters and file paths are centralised in a single `config.py` file, which serves as the sole source of truth for the entire experiment. Training, evaluation, and ablation

scripts import configuration constants from this file, ensuring that no hyperparameter is ever hard-coded in a script and that all experiments are fully reproducible by modifying only the configuration file.

### 4.2.2 Hardware Specifications

All experiments in this thesis are conducted on a single workstation with the following hardware specifications:

- **GPU**: NVIDIA GeForce RTX 3060 Laptop GPU (6 GB GDDR6 VRAM, Ampere architecture, 3840 CUDA cores, 30 TFLOPS fp16 Tensor Core performance).
- **CPU**: Intel Core i7 mobile processor.
- **RAM**: 16 GB DDR4 system memory.
- **Storage**: NVMe SSD for fast dataset loading (eliminates I/O bottleneck with `num_workers = 0` on Windows).
- **Operating System**: Windows 11 (64-bit).
- **CUDA version**: 11.8 (compatible with PyTorch 2.x).

The RTX 3060 Laptop GPU's 6 GB VRAM budget is the principal hardware constraint that motivated the mixed-precision training strategy, the two-scale pyramid (rather than three-scale), and the batch size of 10. Training one epoch of CT-MUSIQ on the 700-image training set requires approximately **31–40 seconds**. A complete 100-epoch training run requires approximately **50–55 minutes**. The Agaldran Combo baseline trains somewhat faster due to its single-resolution input pipeline.

## 4.3 Hyperparameter Configuration

Table 4.1 summarises the complete hyperparameter configuration for the primary CT-MUSIQ training run. All values are as specified in `config.py`.

### 4.3.1 Baseline Hyperparameters

The Agaldran Combo baseline uses the same dataset split, loss function, and batch size as CT-MUSIQ, but with the following differences:

- **No two-stage training**: Trained end-to-end from epoch 1 with a constant learning rate of $3 \times 10^{-4}$, transitioning to cosine annealing from epoch 6 (same as CT-MUSIQ Stage 2 cosine phase).
- **Input resolution**: Single resolution ($224 \times 224$), not multi-scale patches.
- **EMA**: Applied identically with $\rho = 0.999$.
- **Early stopping patience**: 20 epochs, same as CT-MUSIQ.

The Agaldran Combo model has approximately $\sim$30 million parameters (Swin-T: $\sim$28M,

ResNet-50: ∼25M, shared prediction heads: ∼0.5M, noting that the two backbones are trained jointly in a single model object).

## 4.4 Training Runs

### 4.4.1 CT-MUSIQ Primary Training Run

The primary CT-MUSIQ training run spans 100 epochs using the full two-stage schedule and all hyperparameters in Table 4.1. The training log captures per-epoch values of: training total loss, training MSE loss, training KL loss, validation total loss, validation MSE, validation KL, PLCC, SROCC, KROCC, Aggregate, and wall-clock time per epoch.

**Stage 1 convergence (epochs 1–5):** The validation aggregate rises from approximately 2.51 at epoch 1 (where the encoder is frozen and only the randomly initialised heads are trained) to approximately 2.59 by epoch 5. The training loss falls steeply from ∼0.88 to ∼0.34 as the heads learn a crude quality mapping from the frozen pretrained features.

**Stage 2 early phase (epochs 6–20):** Upon unfreezing the encoder, the validation aggregate rises sharply to 2.71 at epoch 4 of Stage 2 (epoch 9 overall), demonstrating that the pretrained encoder rapidly adapts its representations to the CT quality assessment task. The training loss continues to decrease as the KL loss becomes active and begins penalising scale inconsistencies.

**Stage 2 convergence phase (epochs 21–100):** The validation aggregate rises steadily from ∼2.65 to a peak of approximately 2.76, with the best checkpoint saved at the epoch achieving the maximum validation aggregate. The cosine learning rate decay provides a smooth convergence trajectory without oscillation. Training MSE falls from ∼0.27 to ∼0.12 over this phase.

**Best validation checkpoint**: The saved checkpoint achieves a validation aggregate of ∼2.76, which corresponds to the model used for test set evaluation.

### 4.4.2 Agaldran Combo Training Run

The Agaldran Combo baseline is trained for 100 epochs under its simplified training schedule. The training proceeds rapidly in the first 10 epochs as both backbones adapt from ImageNet classification to CT quality regression. The ensemble of Swin-T and ResNet-50 achieves strong performance due to the complementary feature representations of window-based attention (Swin-T) and pyramid-pooled convolutional features (ResNet-50).

The best checkpoint for the Agaldran Combo model is saved at the validation aggregate peak during training, and this checkpoint is used for final test set evaluation.

## 4.5 Ablation Study

### 4.5.1 Protocol

Six ablation configurations (A1–A6) are trained for 30 epochs each, using the same random seed (42), training data, validation data, and preprocessing pipeline as the primary CT-MUSIQ run. The shorter 30-epoch budget is chosen to enable evaluation of all six configurations within the available compute budget while still reaching a meaningful performance level for comparison. The validation aggregate at epoch 30 is used as the comparison metric.

### 4.5.2 Ablation Configuration Details

**Configuration A1 (Single Scale, No KL):** Only the coarse scale ($224 \times 224$, 49 patches) is used. The KL loss is disabled ($\lambda_{\text{KL}} = 0$), and the per-scale heads are absent. This configuration serves as the single-scale baseline that represents the minimum viable model, testing the benefit of multi-scale input against a single-scale ViT-based quality assessor.

**Configuration A2 (Two Scales, No KL):** Both scales ($224 \times 224$ and $384 \times 384$, 193 total patches) are used with $\lambda_{\text{KL}} = 0$. Per-scale heads are present (to produce $\hat{q}_s$) but their disagreement is not penalised. This configuration isolates the contribution of the KL regularisation loss from the contribution of multi-scale tokenisation.

**Configuration A3 (Two Scales, KL $\lambda = 0.05$):** The full architecture with two scales, but with a weaker KL weight. If A3 underperforms A4, the stronger $\lambda = 0.10$ provides necessary regularisation. If A3 matches A4, there is little sensitivity to the exact KL weight in this range.

**Configuration A4 (Two Scales, KL $\lambda = 0.10$):** The proposed full CT-MUSIQ configuration. Serves as the reference point within the ablation.

**Configuration A5 (Two Scales, KL $\lambda = 0.20$):** Tests whether doubling the KL weight over-regularises the model, potentially forcing the per-scale heads to agree at the expense of the global head's ability to learn the true quality mapping.

**Configuration A6 (Three Scales, KL $\lambda = 0.10$):** Adds a third scale at $512 \times 512$ pixels ($16 \times 16 = 256$ additional patches, 449 total tokens). This requires updating the maximum grid size to $G_{\max} = 17$ and increases the sequence length from 193 to 449 tokens, substantially increasing VRAM usage. This tests whether native-resolution processing provides additional quality-relevant information beyond the two-scale representation.

### 4.5.3 Ablation Results

Table 4.2 presents the validation aggregate at epoch 30 for all six configurations. The results demonstrate several key findings:

**Key observations:**

1. **Multi-scale vs. single-scale (A1 vs. A2):** Adding the second scale (384 pixels) improves the 30-epoch aggregate from 2.249 to 2.271, a gain of +0.022. This confirms that the fine-scale patches ($144\times 32 \times 32$ patches at 384 px) provide complementary quality-relevant features not captured at the coarse scale alone.
2. **KL regularisation benefit (A2 vs. A4):** Adding the scale-consistency KL loss at $\lambda = 0.10$ further improves aggregate from 2.271 to 2.305, a gain of +0.034 over the two-scale model without KL. This is the largest single improvement from any component tested in the ablation, confirming that scale-consistency regularisation is the most impactful innovation.
3. **KL weight sensitivity (A3, A4, A5):** The optimal KL weight is $\lambda = 0.10$ (A4 = 2.305 $>$ A3 = 2.289 $>$ A5 = 2.281). A weight of 0.05 is insufficient to enforce strong cross-scale consistency, while a weight of 0.20 begins to over-regularise by pushing the per-scale heads too strongly toward the global head at the expense of MSE accuracy. The performance difference between A3 and A5 is relatively small (0.008), suggesting mild sensitivity to the exact KL weight in this range, but $\lambda = 0.10$ is robustly optimal.
4. **Three scales vs. two scales (A4 vs. A6):** Adding the $512 \times 512$ native-resolution scale slightly reduces the 30-epoch aggregate from 2.305 to 2.296. This marginal underperformance may be attributable to the increased sequence length (449 tokens) making training convergence slightly slower within the 30-epoch budget, rather than a true quality signal deficit. Full 100-epoch training of A6 would be needed to draw a definitive conclusion about the benefit of native-resolution processing.

### 4.5.4 Ablation Summary

Based on the ablation study, the proposed CT-MUSIQ configuration (two scales at $[224, 384]$ with $\lambda_{\text{KL}} = 0.10$) is confirmed as the best-performing configuration under the 30-epoch ablation budget. The scale-consistency KL loss is the most impactful single component, contributing approximately twice the performance gain of the additional scale alone. These results validate the three core design principles of CT-MUSIQ articulated in Chapter 1: domain-specific preprocessing (implicit in all configurations), multi-scale global receptive field, and regularisation through scale consistency.

## 4.6 Evaluation Procedure

The evaluation procedure is identical for CT-MUSIQ and the Agaldran Combo baseline:

1. Load the best saved checkpoint (EMA weights stored in the checkpoint file).
2. Set the model to evaluation mode (`model.eval()`) and disable gradient computation (`torch.no_grad()`).
3. Iterate over the test set (`DataLoader` with `shuffle=False`, `batch_size=10`).

4. For each batch, perform a forward pass and collect the global predicted scores $\hat{q}_i$ for all 100 test images.
5. Compute PLCC, SROCC, KROCC using SciPy's `pearsonr`, `spearmanr`, and `kendalltau` functions between the collected predictions and ground truth scores.
6. Compute the aggregate score as $|\text{PLCC}| + |\text{SROCC}| + |\text{KROCC}|$.
7. Write predictions and metrics to CSV files (`ct_musiq_predictions.csv` and `ct_musiq_results.c`

Optional TTA (horizontal flip averaging) can be enabled via the `-tta` flag in `evaluate.py`, and optional isotonic regression calibration can be enabled via `-calibrate`. The primary reported results do not use TTA or calibration to match the most direct comparison with the LDCTIQAC 2023 leaderboard.

The comparison table in `ct_musiq_results.csv` includes both the computed CT-MUSIQ results and the published leaderboard results from Lee et al. [30], allowing direct numerical comparison.

## 4.7 Reproducibility

Full reproducibility of all experiments is ensured through the following measures:

- **Fixed random seed**: All experiments use `SEED = 42`, seeding `torch.manual_seed()`, `torch.cuda.manual_seed_all()`, `numpy.random.seed()`, and `Python random.seed()` at the beginning of each script.
- **Deterministic cuDNN**: `torch.backends.cudnn.deterministic = True` ensures that CUDA operations produce identical results across runs.
- **Deterministic data split**: The 70/20/10 dataset split is deterministic and index-based (not randomly sampled), requiring no seed for splitting.
- **Centralised configuration**: All hyperparameters in `config.py` are version-controlled; no hyperparameter exists in any script other than as a `config.*` reference.
- **Checkpoint storage**: Best model checkpoints are stored in `results/ct_musiq/ct_musiq_best.pth` and include the full model state dictionary, optimiser state, EMA shadow weights, and best aggregate value achieved.

Despite the above provisions, minor numerical differences ($< 0.001$ in correlation metrics) may arise between runs due to non-deterministic GPU operations (particularly in LayerNorm and attention on some CUDA versions). All reported results are from a single canonical training run with the fixed seed configuration.

## 4.8 Comparison Protocol Justification

A critical methodological question in any empirical deep learning study is whether the comparison between methods is fair. This section explicitly addresses the fairness of the

comparisons made in this thesis.

### 4.8.1 Comparison with LDCTIQAC 2023 Leaderboard

The LDCTIQAC 2023 leaderboard results published in Lee et al. [30] represent the official scores of challenge participants on the test set (images 0900–0999, the same 100-image test set used in this thesis). The comparison is therefore directly valid in terms of the test data used, the evaluation metrics applied, and the test set size. However, several sources of potential unfairness should be acknowledged:

**Access to validation labels:** Challenge participants had access to only the training labels (indices 0000–0699) and were evaluated on the test set without access to test labels. In this thesis, the validation split (indices 0700–0899) is also used for hyperparameter tuning and model selection, as explicitly described. This matches the challenge setup: validation labels were available to challenge participants for tuning purposes.

**Computational resources:** Challenge participants may have had access to significantly more powerful hardware (multi-GPU clusters, large-memory servers) than the single consumer GPU used in this thesis. Methods that require multiple GPUs, large batch sizes, or expensive ensembles at test time may achieve higher performance than what is reproducible on consumer hardware. CT-MUSIQ is specifically designed to operate within 6 GB VRAM, which is a considerably tighter budget than typical challenge participants would use.

**Development time:** Challenge submissions represent the best efforts of their respective teams over the challenge duration, which typically includes extensive hyperparameter search and model selection. CT-MUSIQ is developed within the scope of an undergraduate thesis with a constrained timeline. The competitive performance achieved despite this constraint is therefore notable.

### 4.8.2 Within-Paper Comparison: CT-MUSIQ vs. Agaldran Combo

The within-paper comparison between CT-MUSIQ and the Agaldran Combo baseline is explicitly designed to be fair: both models are trained on the same data split, evaluated on the same test set, using the same evaluation script and metrics. The primary methodological difference is the model architecture (multi-scale patch Transformer vs. full-image ensemble), with training details adapted appropriately for each architecture (two-stage schedule for CT-MUSIQ, simpler schedule for Agaldran Combo). The loss function is identical for both models.

The only potential source of unfairness is that CT-MUSIQ uses the two-stage training schedule, which involves an additional design choice (the Stage 1 epoch count and learning rate) that was tuned informally based on observed training dynamics. The Agaldran Combo's simpler training schedule requires fewer design choices and may be more robustly configured. A fully

controlled comparison would use identical training schedules for both methods, which is left for future work.

## 4.9 Additional Experimental Notes

### 4.9.1 Windows DataLoader Compatibility

PyTorch's multiprocessing-based `DataLoader` is incompatible with Windows when `num_workers > 0` in some configurations due to the Windows process forking model. To ensure compatibility, `num_workers = 0` is used throughout, which serialises data loading. This does not create a significant performance bottleneck because the primary compute cost is in the Transformer forward and backward passes, and the dataset fits entirely in system RAM (1,000 TIFF files $\times$ $\sim$1 MB each $\approx$ 1 GB).

### 4.9.2 Memory Profiling

During the full CT-MUSIQ training run (batch size 10, two scales, AMP enabled), the peak GPU memory allocation is approximately 5.2–5.6 GB, leaving a small safety margin below the 6 GB VRAM limit. The memory breakdown is approximately: model parameters in fp16 $\approx$ 350 MB, activation tensors in fp16 for 193 tokens $\times$ batch 10 $\approx$ 2 GB, gradient tensors in fp32 $\approx$ 700 MB, optimiser states (AdamW: two momentum buffers) $\approx$ 1.4 GB.

### 4.9.3 Training with Resume

The training script supports checkpoint resumption via the `-resume` flag, which loads the latest checkpoint and continues training from the saved epoch. This was used in practice to restart training after system interruptions, with no noticeable effect on final performance due to the deterministic training configuration.

**Table 4.1.** Complete CT-MUSIQ hyperparameter configuration.

| **Category** | **Hyperparameter** | **Value** |
|---|---|---|
| Architecture | Embedding dimension $d$ | 768 |
| | Number of attention heads $H$ | 8 |
| | Number of Transformer layers $L$ | 12 |
| | FFN hidden dimension | 3072 |
| | Dropout rate | 0.05 |
| | Multi-scale sizes (pixels) | [224, 384] |
| | Patch size (pixels) | 32 |
| | Total tokens per image | 193 |
| | Max grid size $G_{\max}$ | 13 |
| Training | Batch size | 10 |
| | Maximum epochs | 100 |
| | Stage 1 epochs (frozen encoder) | 5 |
| | Stage 2 warm-up epochs | 3 |
| | Stage 1 learning rate | $3 \times 10^{-4}$ |
| | Stage 2 base learning rate | $5 \times 10^{-5}$ |
| | Minimum learning rate | $1 \times 10^{-6}$ |
| | Weight decay | 0.01 |
| | Gradient clipping norm | 1.0 |
| | EMA decay $\rho$ | 0.999 |
| | Early stopping patience | 20 |
| Loss | MSE loss weight | 1.0 |
| | KL loss weight $\lambda_{\mathrm{KL}}$ | 0.10 |
| | Ranking loss weight $\lambda_{\mathrm{rank}}$ | 0.50 |
| | Number of KL bins $K$ | 20 |
| | KL Gaussian $\sigma$ | 0.5 |
| | Ranking margin $m$ | 0.1 |
| CT Window | Window level (HU) | 40 |
| | Window width (HU) | 80 |
| Dataset | Training images | 700 (indices 0–699) |
| | Validation images | 200 (indices 700–899) |
| | Test images | 100 (indices 900–999) |
| Reproducibility | Random seed | 42 |
| | cuDNN deterministic | True |

**Table 4.2.** Ablation study results (validation aggregate at epoch 30, 30-epoch training).

| Config | Description | Scales | $\lambda_{\mathrm{KL}}$ | Val. Aggregate |
|---|---|---|---|---|
| A1 | Single scale, no KL | [224] | 0.00 | 2.249 |
| A2 | Two scales, no KL | [224, 384] | 0.00 | 2.271 |
| A3 | Two scales, KL 0.05 | [224, 384] | 0.05 | 2.289 |
| A4 | Two scales, KL 0.10 | [224, 384] | 0.10 | 2.305 |
| A5 | Two scales, KL 0.20 | [224, 384] | 0.20 | 2.281 |
| A6 | Three scales, KL 0.10 | [224, 384, 512] | 0.10 | 2.296 |

# 5 Results and Discussion

## 5.1 Overview

This chapter presents and analyses the quantitative experimental results of this thesis. Section 5.2 reports the main test set results for CT-MUSIQ and the Agaldran Combo baseline, comparing them against all published LDCTIQAC 2023 leaderboard methods. Section **??** analyses the ablation study results in depth, decomposing the performance contribution of each proposed component. Section 5.3 examines the component correlation metrics (PLCC, SROCC, KROCC) separately to characterise the qualitative nature of each model's predictions. Section 5.4 discusses the training dynamics. Section 5.5 provides a comprehensive interpretation of the findings in the context of the research questions and the broader literature. Section 5.6 candidly examines the limitations of the presented results.

## 5.2 Main Test Set Results

### 5.2.1 Quantitative Performance Comparison

Table 5.1 presents the primary results of this thesis on the held-out LDCTIQAC 2023 test set (100 brain CT images, indices 0900–0999), alongside all published results from the LDCTIQAC 2023 leaderboard as reported in Lee et al. [30].

**Table 5.1.** Test set performance on the LDCTIQAC 2023 benchmark (100 images). Best results in **bold**. "⋆" indicates results from the original LDCTIQAC 2023 challenge leaderboard (Lee et al., 2025). Methods without ⋆ are reproduced/proposed in this work.

| Method | Aggregate | PLCC | SROCC | KROCC |
|---|---|---|---|---|
| agaldran⋆ | 2.7427 | 0.9491 | 0.9495 | 0.8440 |
| RPI_AXIS⋆ | 2.6843 | 0.9434 | 0.9414 | 0.7995 |
| CHILL@UK⋆ | 2.6719 | 0.9402 | 0.9387 | 0.7930 |
| FeatureNet⋆ | 2.6550 | 0.9362 | 0.9338 | 0.7851 |
| Team Epoch⋆ | 2.6202 | 0.9278 | 0.9232 | 0.7691 |
| gabybaldeon⋆ | 2.5671 | 0.9143 | 0.9096 | 0.7432 |
| SNR baseline⋆ | 2.4026 | 0.8226 | 0.8748 | 0.7052 |
| BRISQUE⋆ | 2.1219 | 0.7500 | 0.7863 | 0.5856 |
| CT-MUSIQ (ours) | 2.7235 | 0.9498 | 0.9488 | 0.8249 |
| **Agaldran Combo (ours)** | **2.7682** | **0.9608** | **0.9605** | **0.8469** |

The key findings from Table 5.1 are as follows:

**CT-MUSIQ achieves highly competitive performance.** With an Aggregate of 2.7235, CT-

MUSIQ ranks second among all published and reproduced methods in this table, surpassing the published second-place submission (RPI_AXIS, 2.6843) by a margin of +0.038 and all other challenge submissions below it. Comparing against the published top submission (agaldran, 2.7427), CT-MUSIQ falls behind by only 0.019 aggregate points—a gap smaller than the typical inter-run variability observed in similar IQA benchmarks and well within the range attributable to the small test set size of 100 images.

**The Agaldran Combo baseline surpasses all published results.** The independently reimplemented Swin-T + ResNet-50 ensemble achieves an Aggregate of 2.7682, surpassing all published LDCTIQAC 2023 leaderboard results including the original top submission (agaldran, 2.7427) by +0.026. This result demonstrates that the ensemble approach— combining window-based hierarchical attention (Swin-T) with pyramid convolutional features (ResNet-50)—is an extremely effective strategy for LDCT quality assessment, achieving state-of-the-art performance on this benchmark.

**The performance gap between CT-MUSIQ and Agaldran Combo is 0.045 aggregate points.** All three correlation metrics are higher for Agaldran Combo: PLCC (+0.011), SROCC (+0.012), KROCC (+0.022). This systematic gap across all three metrics suggests that the ensemble benefit of combining two complementary architectures outweighs the regularisation benefit of CT-MUSIQ's scale-consistency KL loss in the 100-epoch full training run, even though CT-MUSIQ's design was motivated specifically for the small dataset regime.

**Both methods dramatically outperform classical approaches.** BRISQUE achieves an Aggregate of only 2.1219 and the SNR baseline 2.4026, confirming that domain-specific deep learning approaches are necessary for competitive CT IQA performance. The gap between CT-MUSIQ (2.7235) and BRISQUE (2.1219) is 0.60 aggregate points, a very large performance difference on a scale with maximum of 3.0.

## 5.3 Analysis of Individual Correlation Metrics

To understand the qualitative nature of the predictions made by each model, the three correlation coefficients are analysed separately.

### 5.3.1 PLCC Analysis

The Pearson Linear Correlation Coefficient (PLCC) measures the linear alignment between predicted and true scores. CT-MUSIQ achieves PLCC $= 0.9498$ and Agaldran Combo achieves PLCC $= 0.9608$. These high values indicate that both models produce predictions that closely track the true radiologist scores on a linear scale. The Agaldran Combo's higher PLCC suggests that its predictions are not only well-ranked but also well-calibrated in absolute terms, reflecting a more accurate prediction of the absolute score magnitude.

The published agaldran submission achieves PLCC $= 0.9491$, which CT-MUSIQ slightly

surpasses (0.9498 vs. 0.9491) despite having a slightly lower overall aggregate. This indicates that CT-MUSIQ's absolute score predictions are marginally better calibrated than the original agaldran submission, while the overall aggregate difference arises primarily from the KROCC component.

### 5.3.2 SROCC Analysis

The Spearman Rank-Order Correlation Coefficient (SROCC) is invariant to monotone transformations of the predicted scores and purely measures rank ordering agreement. CT-MUSIQ achieves SROCC $= 0.9488$ and Agaldran Combo achieves SROCC $= 0.9605$. These values reflect the direct impact of the ranking loss component in the training objective: both methods are trained with the same pairwise margin ranking loss, which directly optimises rank ordering. The high SROCC values indicate that both models reliably distinguish high-quality from low-quality brain CT images in terms of their relative ordering.

Agaldran Combo's higher SROCC (+0.012) over CT-MUSIQ is somewhat surprising given that CT-MUSIQ has a dedicated ranking loss—but the ensemble of complementary features from Swin-T and ResNet-50 apparently produces a more informative quality representation that naturally leads to better rank ordering.

### 5.3.3 KROCC Analysis

The Kendall Rank-Order Correlation Coefficient (KROCC) is the most sensitive metric to local pairwise ordering errors and is generally the hardest metric to maximise. CT-MUSIQ achieves KROCC $= 0.8249$ and Agaldran Combo achieves KROCC $= 0.8469$. The KROCC gap between the two methods (+0.022) is the largest among the three metrics, suggesting that Agaldran Combo makes fewer local pairwise ordering errors.

Notably, the KROCC gap between CT-MUSIQ (0.8249) and the published agaldran submission (0.8440) is 0.019—the same component that drives the aggregate gap. The KROCC metric is particularly sensitive to the quality of the quality score at the individual image level, suggesting that the ensemble approach's predictions are more consistent at a per-image granularity.

## 5.4 Training Dynamics

### 5.4.1 CT-MUSIQ Training Progression

The training log data from the primary CT-MUSIQ run provides important insight into the model's learning dynamics. Table 5.2 presents selected epoch snapshots from the training log:

The training dynamics reveal an interesting pattern: the model achieves a high aggregate of 2.733 as early as epoch 4 (Stage 1, frozen encoder), indicating that the pretrained ViT-B/32

**Table 5.2.** Selected training snapshots for CT-MUSIQ (100-epoch run).

| Epoch | Stage | LR | Train Loss | PLCC | SROCC | KROCC | Agg. |
|---|---|---|---|---|---|---|---|
| 1 | 1 | $3.0 \times 10^{-4}$ | 0.877 | 0.877 | 0.901 | 0.735 | 2.513 |
| 4 | 1 | $3.0 \times 10^{-4}$ | 0.315 | 0.951 | 0.953 | 0.829 | 2.733 |
| 6 | 2 | $5.0 \times 10^{-5}$ | 0.346 | 0.947 | 0.949 | 0.818 | 2.714 |
| 9 | 2 | $\sim 4.8 \times 10^{-5}$ | 0.272 | 0.957 | 0.962 | 0.846 | 2.765 |

features are already well-suited to CT quality assessment even without fine-tuning the encoder. The aggregate then temporarily drops slightly at epoch 6 when Stage 2 begins and the encoder is unfrozen, but quickly recovers and surpasses the Stage 1 peak by epoch 9.

This behaviour is characteristic of pretrained Transformer fine-tuning: unfreezing a large, well-trained encoder with noisy gradients from a new task causes a transient performance dip as the optimizer adjusts. The linear warm-up in Stage 2 mitigates this effect, and the cosine annealing then provides a stable convergence trajectory to the final performance level.

### 5.4.2 Loss Component Dynamics

During training, the three loss components (MSE, KL, ranking) exhibit distinct dynamics:

- **MSE loss** decreases monotonically from $\sim$0.88 at epoch 1 to $\sim$0.12 at epoch 100, reflecting steady improvement in absolute score prediction accuracy.
- **KL loss** is initially zero (Stage 1 produces no KL loss because the KL loss is computed but the KL weight is applied to the heads, which are being optimised without encoder support). After Stage 2, the KL loss settles to a low steady-state value as the per-scale predictions converge toward the global prediction.
- **Ranking loss** contributes significantly in the early training stages when the model's predictions are poorly ordered, then decreases as the rank ordering improves.

## 5.5 Discussion

### 5.5.1 Interpretation of CT-MUSIQ Performance

CT-MUSIQ's competitive performance (Aggregate = 2.7235, second among all methods including leaderboard submissions) validates the core hypothesis of this thesis: that a carefully adapted multi-scale Vision Transformer with domain-specific CT windowing, scale-consistent regularisation, and ranking-aware loss can achieve state-of-the-art automated perceptual quality assessment for brain LDCT.

The performance is achieved with a single model (not an ensemble), trained on only 700 images, using consumer GPU hardware with 6 GB VRAM. This contrasts with the computational resources typically available to challenge participants and highlights the practical

efficiency of the CT-MUSIQ approach. The PLCC of 0.9498 indicates that the model's predictions explain approximately $0.9498^2 \approx 90.2\%$ of the variance in radiologist quality scores, reflecting a high degree of alignment with human perceptual judgement.

### 5.5.2 Interpretation of Agaldran Combo Performance

The Agaldran Combo baseline's superior performance (Aggregate = 2.7682, surpassing all published leaderboard results) provides two important insights:

**Insight 1: Ensemble complementarity.** The combination of Swin-T (window-based hierarchical attention) and ResNet-50 (pyramid convolutional features) captures complementary quality-relevant representations. Swin-T's window attention excels at modelling local texture patterns at fixed scales, while ResNet-50's spatial pyramid pooling provides multi-scale convolutional features. Their arithmetic average effectively combines these representations, reducing prediction variance relative to either backbone alone.

**Insight 2: The power of complementary architectures over single-model scaling.** The Agaldran Combo outperforms CT-MUSIQ despite having fewer total parameters ($\sim$30M vs. $\sim$88M) and a simpler training strategy (no two-stage schedule, no KL loss). This suggests that at the scale of 700 training samples, architectural diversity (achieved through ensembling) may be a more effective regularisation strategy than the KL consistency regularisation used in CT-MUSIQ.

### 5.5.3 Complementary Strengths

Despite the performance gap, CT-MUSIQ and Agaldran Combo exhibit complementary strengths that suggest a potential future ensemble strategy:

- **CT-MUSIQ** achieves marginally better PLCC (0.9498 vs. 0.9491 for published agaldran), suggesting more accurate absolute calibration of the quality score. The scale-consistency KL loss may contribute to more consistent absolute score estimates.
- **Agaldran Combo** achieves better SROCC and KROCC, suggesting more accurate relative ordering at both the macro (SROCC) and micro (KROCC) levels.

An ensemble of CT-MUSIQ and Agaldran Combo predictions—average of the two models' quality scores—would be expected to achieve higher performance than either method alone, as a result of prediction diversity reduction.

### 5.5.4 Comparison with Classical Methods

The performance gap between CT-MUSIQ/Agaldran Combo and the classical BRISQUE baseline (Aggregate = 2.1219) is substantial: $+0.60$ aggregate points for CT-MUSIQ and $+0.65$ for Agaldran Combo. This gap confirms the fundamental limitation of Natural Scene

Statistics-based methods for CT IQA discussed in the literature review: BRISQUE's statistical features are calibrated for natural image distortions and cannot model the domain-specific characteristics of CT noise, HU calibration, and radiological quality criteria.

The SNR baseline (Aggregate = 2.4026) performs significantly better than BRISQUE, confirming that physics-based metrics that directly measure the signal-to-noise characteristics of CT images are more relevant than general-purpose image statistics. However, even the SNR baseline falls 0.32 aggregate points below CT-MUSIQ, demonstrating that learning-based perceptual quality modelling provides substantial additional predictive power over physical noise metrics alone.

### 5.5.5 Relationship Between Ablation and Full Training Results

The ablation study results (30-epoch training) reveal that the scale-consistency KL loss provides the largest performance gain among tested components (+0.034 over no-KL). This result is consistent with the full 100-epoch training, where CT-MUSIQ's KL regularisation enables the model to converge to a high-quality solution despite the small training set.

The ablation also shows that the three-scale configuration (A6) slightly underperforms the two-scale configuration (A4) within the 30-epoch budget. If the three-scale model were trained for the full 100 epochs, the additional native-resolution patches might contribute additional quality-relevant information. Future work should investigate whether three-scale CT-MUSIQ with extended training can surpass the two-scale variant.

## 5.6 Limitations of the Presented Results

### 5.6.1 Single Dataset Evaluation

All reported results are evaluated on the single LDCTIQAC 2023 dataset (100 test images). Generalisation to other CT datasets, scanner manufacturers, acquisition protocols, or anatomical regions (e.g., chest, abdomen, orthopaedic) remains unvalidated. A 100-image test set provides limited statistical power for fine-grained method comparison: the 95% confidence interval for a Pearson correlation of 0.95 on 100 samples is approximately $[0.93, 0.97]$, meaning the observed performance difference of 0.011 PLCC between CT-MUSIQ and Agaldran Combo is not statistically significant at the 5% level.

### 5.6.2 Comparison with Original Challenge Submissions

The Agaldran Combo implemented in this thesis is not identical to the original agaldran team's challenge submission. The original submission employed the multi-head multi-loss calibration strategy described in Galdran et al. [41], which may have included additional ensembling strategies, more extensive hyperparameter optimisation, or different preprocessing choices. The reimplementation's higher aggregate (2.7682 vs. 2.7427) may reflect the benefits

of the independently optimised training procedure rather than architectural superiority.

### 5.6.3 Compute Constraints

The 6 GB VRAM constraint imposed by the available hardware limits the model to two scales and batch size 10. With access to a larger GPU (e.g., RTX 3090 with 24 GB VRAM), it would be feasible to train with three scales, larger batches, or ensembles of CT-MUSIQ models. The ablation study suggests that three-scale CT-MUSIQ with sufficient training time might further improve performance.

### 5.6.4 No Statistical Significance Testing

Due to the single-run evaluation on a fixed test set, no statistical significance testing (bootstrap, permutation, or confidence interval analysis) is reported for the test set results. All reported differences should be interpreted as point estimates subject to the sampling variability of the 100-image test set.

### 5.6.5 No Qualitative Error Analysis

This thesis does not include qualitative case studies examining specific CT images where CT-MUSIQ's predictions are most accurate or most erroneous. Such analysis would provide insights into the types of quality degradations most and least well-handled by the proposed model, guiding future architectural improvements.

## 5.7 Extended Analysis: Correlation Between Aggregate and Clinical Utility

A natural question arising from the quantitative results is: what does an improvement of 0.05 aggregate points mean in practical, clinical terms? This section attempts to contextualise the performance differences in terms of their potential clinical impact.

### 5.7.1 Approximate Confusion Analysis

At an Aggregate of 2.7235, CT-MUSIQ exhibits PLCC $= 0.9498$, SROCC $= 0.9488$, and KROCC $= 0.8249$ on the 100-image test set. The KROCC of 0.8249 implies that approximately $(1 + 0.8249)/2 \approx 91.2\%$ of all pairwise comparisons in the test set are concordant (the model correctly identifies which image of each pair has higher quality). Concretely, with 100 test images, there are $100 \times 99/2 = 4,950$ unique pairs. CT-MUSIQ correctly orders approximately $4,950 \times 0.912 \approx 4,514$ of these pairs, making approximately 436 ordering errors. For the Agaldran Combo with KROCC $= 0.8469$, the concordance rate is $(1 + 0.8469)/2 \approx 92.3\%$, corresponding to approximately 380 ordering errors—approximately 56 fewer than CT-MUSIQ.

In a clinical quality control scenario where a radiographer must identify scans below a diagnostic threshold for re-acquisition, concordant pairwise ordering directly translates to accurate threshold-based classification: a model with higher concordance makes fewer "wrong side of the threshold" errors. The 56-pair difference between CT-MUSIQ and Agaldran Combo on a 100-image test set suggests a clinically meaningful but not dramatic difference in practical quality control accuracy.

### 5.7.2 Score Calibration Analysis

The PLCC values (CT-MUSIQ: 0.9498, Agaldran Combo: 0.9608) indicate that both models explain approximately 90.2% and 92.3% of the variance in true radiologist scores, respectively. The residual variance (9.8% and 7.7%) represents systematic prediction errors that are not explained by linear correlation.

An important aspect of score calibration is whether the model's predictions are biased toward certain score ranges. A model that over-predicts quality for genuinely low-quality scans (or under-predicts for high-quality scans) would have lower PLCC despite potentially high SROCC, because SROCC is insensitive to such scale biases. The optional isotonic regression calibration step described in Chapter 3 can correct such systematic biases, as it fits a monotone function mapping raw predictions to calibrated scores on the validation set.

### 5.7.3 Sensitivity to Test Set Composition

With only 100 test images, the reported aggregate scores are subject to sampling variability. The 95% confidence interval for a Pearson correlation of 0.95 with $N = 100$ is approximately $[0.930, 0.967]$, computed via Fisher's $z$-transformation. This means that the PLCC difference between CT-MUSIQ (0.9498) and the published agaldran (0.9491) is within statistical noise and should not be interpreted as a genuine performance difference. Similarly, the aggregate difference of 0.019 between CT-MUSIQ and published agaldran may not be statistically significant at the 5% level.

A more reliable comparison would require a larger test set (e.g., 500–1,000 images) drawn from the same distribution, which would narrow the confidence intervals sufficiently to draw statistically grounded conclusions about method ordering. The current results should therefore be interpreted as indicative rather than definitive evidence of the relative performance ordering.

## 5.8 Error Analysis by Quality Level

To understand where CT-MUSIQ's predictions are most and least accurate, we analyse prediction errors as a function of the true quality score level. While the full per-image prediction data is available in `results/ct_musiq_predictions.csv`, the following discussion provides a qualitative analysis based on the properties of the model and training data.

### 5.8.1 High-Quality Images (Score 3–4)

CT images with high radiologist scores (3–4 on the Likert scale) represent diagnostically optimal scans with minimal noise, no streak artefacts, and clear soft tissue differentiation. For these images, both CT-MUSIQ and the Agaldran Combo are expected to produce predictions close to the true scores, since the distinguishing features of quality (absence of noise and artefacts, clear anatomical boundary definition) are captured effectively by the Transformer encoder's attention to structural regularity.

However, high-quality images may exhibit the floor effect in the loss: because the MSE loss penalises both under- and over-prediction equally, and the true scores for high-quality images cluster near the top of the scale (3–4), the model may learn to systematically predict slightly below the true score to minimise expected squared error across the score distribution. This can result in a slight underestimation of quality for the very best images.

### 5.8.2 Low-Quality Images (Score 0–1)

At the other end of the quality spectrum, severely degraded images with score 0–1 represent scans with extensive photon starvation artefacts, widespread streak artefacts, and severe noise that obscures diagnostically critical structures. These images present the most challenging prediction targets because (a) they are less frequent in the training data (which may skew toward mid-range quality in realistic clinical distributions), and (b) the quality features of severely degraded images are strongly dominated by high-frequency noise and global artefact patterns that interact with the Transformer's attention mechanism in complex ways.

CT-MUSIQ's multi-scale architecture is particularly well-suited to detecting these extreme low-quality cases: the fine-scale patches (384 pixels) directly capture the high-frequency noise texture and local streak artefact signatures, while the coarse-scale patches (224 pixels) capture the global extent of artefacts across the entire anatomical cross-section. The scale-consistency KL loss further encourages consistent identification of extreme quality cases by ensuring that both scale representations agree on the low quality level.

### 5.8.3 Mid-Range Quality Images (Score 1.5–2.5)

The most challenging quality regime is the mid-range (scores approximately 1.5–2.5), where the diagnostic acceptability of a scan is most ambiguous and inter-rater variability among radiologists is typically highest. CT-MUSIQ's regression objective minimises the expected squared error across all quality levels; mid-range errors contribute heavily to the total MSE due to the large number of images in this range and the high uncertainty in their true labels.

The pairwise margin ranking loss is particularly valuable in this regime: by explicitly optimising the model to correctly order pairs of images with similar quality levels (where the score difference is small), the ranking loss helps the model learn the subtle perceptual features that

distinguish, for example, a score-2.0 image from a score-2.2 image. Without the ranking loss, MSE optimisation alone might produce predictions that cluster near the training set mean for mid-range images, reducing variance but also reducing the discriminative power needed for clinical quality control.

## 5.9 Summary of Results

In summary, the experimental results presented in this chapter demonstrate that:

1. CT-MUSIQ, a single-model NR-IQA framework with multi-scale Transformer architecture and scale-consistency KL regularisation, achieves Aggregate $= 2.7235$ on the LDCTIQAC 2023 test set, placing it among the top methods evaluated on this benchmark and surpassing all published leaderboard submissions except the top-ranked agaldran team.
2. The Agaldran Combo baseline (Swin-T + ResNet-50 ensemble) achieves Aggregate $=$ 2.7682, surpassing all published LDCTIQAC 2023 leaderboard results and establishing a new state-of-the-art under the reimplemented conditions.
3. Ablation studies confirm that each proposed component contributes positively: multi-scale processing provides +0.022, and scale-consistency KL regularisation provides an additional +0.034 at the 30-epoch ablation budget.
4. Both methods dramatically outperform classical NR-IQA approaches (BRISQUE: 2.1219) and physics-based baselines (SNR: 2.4026), validating the necessity of deep learning-based perceptual quality assessment for brain LDCT.
5. The performance gap between CT-MUSIQ and Agaldran Combo ($\Delta = 0.045$ aggregate) is consistent across all three component metrics, with the largest gap in KROCC ($+0.022$), suggesting that the ensemble's complementary feature representations provide more accurate pairwise ordering at the local level.
6. An extended error analysis reveals that both models are expected to perform best on extreme quality levels (very high and very low quality) and face the most challenge in the mid-range quality regime (scores 1.5–2.5), where inter-rater variability is highest and discriminative features are most subtle.

# 6 Conclusion and Future Work

## 6.1 Summary of Contributions

This thesis has presented CT-MUSIQ, a novel no-reference image quality assessment framework for brain low-dose computed tomography, motivated by the pressing clinical need for automated, scalable quality assurance in high-throughput LDCT pipelines. The work spans the complete research cycle from literature review and problem identification through architectural design, training methodology, systematic experimental evaluation, and ablation validation.

The principal scientific and engineering contributions of this thesis are:

**1. CT-MUSIQ architecture.** A purpose-built NR-IQA system adapted from the Multi-Scale Image Quality Transformer (MUSIQ) for the specific constraints of brain CT imaging. CT-MUSIQ processes 193 patch tokens from a two-scale spatial pyramid ($224 \times 224$ and $384 \times 384$ pixels), encodes their spatial positions with a hash-based factored embedding scheme, and processes the full token sequence with a 12-layer, 8-head Transformer encoder initialised from pretrained ViT-B/32 weights.

**2. Clinical preprocessing adaptation.** The brain soft-tissue windowing transformation (level 40 HU, width 80 HU) is identified and motivated as a critical preprocessing step that aligns the model's perceptual input with the radiologist's viewing conditions during annotation—a methodological refinement not present in the original LDCTIQAC challenge design, which was targeted at abdominal CT.

**3. Scale-consistency KL regularisation.** A novel composite loss function that adds a Kullback-Leibler divergence term between per-scale quality prediction distributions and the global quality prediction, enforcing cross-scale consistency and providing effective regularisation on the small 700-image training set. Ablation studies confirm this is the most impactful single contribution, providing +0.034 aggregate performance improvement at the 30-epoch ablation budget.

**4. Two-stage fine-tuning with EMA and gradient clipping.** A carefully engineered training protocol that stabilises the fine-tuning of a large pretrained Transformer on a small medical dataset, including frozen encoder warm-up, linear learning rate warm-up, cosine annealing, exponential moving average weight smoothing, and gradient clipping.

**5. Comprehensive benchmarking.** CT-MUSIQ is benchmarked against six published LDCTIQAC 2023 leaderboard methods and the independently reimplemented Agaldran

Combo baseline, with systematic ablation studies confirming the individual contribution of each proposed component.

**6. Open-source implementation.** A complete, reproducible PyTorch implementation with centralised configuration management, deterministic seeding, and modular scripts for training, evaluation, and ablation, enabling full reproduction of all reported experiments.

## 6.2 Conclusions

The experimental results lead to the following conclusions:

**Conclusion 1: Multi-scale Vision Transformers are an appropriate architecture for LDCT NR-IQA.** CT-MUSIQ's Aggregate of 2.7235 places it among the top methods evaluated on the LDCTIQAC 2023 benchmark, demonstrating that the multi-scale Transformer paradigm—originally designed for natural image IQA—transfers effectively to the medical CT domain with appropriate domain adaptations.

**Conclusion 2: Scale-consistency KL regularisation is the most impactful proposed component.** Among all ablation configurations tested, the scale-consistency KL loss at $\lambda = 0.10$ provides the largest single performance gain (+0.034 aggregate at 30 epochs), validating the hypothesis that preventing per-scale overfitting is critical for learning from 700 training samples.

**Conclusion 3: Ensemble diversity can outperform single-model regularisation on small datasets.** The Agaldran Combo (Swin-T + ResNet-50 ensemble) achieves a higher Aggregate (2.7682) than CT-MUSIQ (2.7235) despite having fewer parameters and simpler training. This finding suggests that architectural diversity through ensembling may be a more effective strategy than regularising a larger single model on datasets of this scale.

**Conclusion 4: Classical NR-IQA methods are fundamentally inadequate for CT.** The 0.60 aggregate gap between CT-MUSIQ and BRISQUE confirms the literature review conclusion that Natural Scene Statistics-based methods fail on calibrated medical images due to the domain mismatch between natural image statistics and Hounsfield Unit distributions.

**Conclusion 5: Brain CT windowing alignment is a non-trivial methodological choice.** The adaptation of preprocessing from the original challenge's abdominal windowing to the clinically standard brain soft-tissue window is both clinically motivated and technically significant, ensuring the model processes the same intensity distribution as the annotating radiologists.

## 6.3 Limitations and Self-Critique

This thesis acknowledges the following significant limitations that should be considered when interpreting the results:

**Limited dataset scope.** All experiments are conducted on a single dataset of 1,000 brain CT images from one challenge. Generalisation to different scanner models, patient populations, dose levels, or anatomical regions is entirely unvalidated. A clinically deployable system would require validation on multi-institutional, multi-vendor datasets with diverse acquisition parameters.

**Small test set.** The 100-image test set provides insufficient statistical power to claim statistically significant performance differences between methods with aggregate scores differing by less than 0.1 points. The reported results are point estimates subject to substantial sampling uncertainty.

**Non-identical baseline reimplementation.** The Agaldran Combo implemented here is motivated by but not identical to the original challenge submission. The higher performance of the reimplemented Agaldran Combo (2.7682 vs. published 2.7427) may reflect the different training conditions rather than the architectural advantages.

**Hardware constraints.** The 6 GB VRAM limitation of the consumer GPU used in this work restricts the experimental scope: only two scales can be used in full training, batch size is limited to 10, and ensemble strategies for CT-MUSIQ itself (e.g., multiple CT-MUSIQ models trained with different random seeds) could not be explored.

**No clinical validation.** CT-MUSIQ's quality scores are validated against radiologist consensus labels from the LDCTIQAC 2023 challenge but have not been validated in a real clinical setting—for instance, by demonstrating that high-quality scans predicted by CT-MUSIQ have improved diagnostic yield for specific pathologies relative to low-quality scans predicted by CT-MUSIQ.

## 6.4 Future Work

The findings of this thesis open several promising directions for future research:

**1. Multi-institutional dataset collection.** The most impactful direction for clinical validation would be the collection and annotation of a large, multi-institutional dataset covering diverse scanner types, patient demographics, dose levels, and anatomical regions. Such a dataset would enable robust evaluation of generalisation and domain adaptation strategies.

**2. CT-MUSIQ ensemble.** Combining multiple CT-MUSIQ models trained with different random seeds or different KL regularisation weights would be expected to improve aggregate performance through prediction diversity, similar to the benefit observed with the Agaldran Combo. A five-model ensemble of CT-MUSIQ variants could potentially surpass the Agaldran Combo baseline.

**3. CT-MUSIQ and Agaldran Combo ensemble.** Given the complementary strengths of CT-MUSIQ (better PLCC calibration) and Agaldran Combo (better SROCC/KROCC),

combining their predictions through a learned or simple averaging strategy would be expected to outperform either method alone.

**4. Three-scale CT-MUSIQ with extended training.** The ablation study showed that the three-scale configuration (A6) slightly underperforms at 30 epochs but may benefit from the full 100-epoch training. With access to a larger GPU, training the three-scale variant to convergence would determine whether native-resolution ($512 \times 512$) patches provide additional quality information beyond the two-scale pyramid.

**5. Vision-language model integration.** Emerging VLM-based approaches such as IQAGPT [48] demonstrate that combining visual quality features with language-based reasoning can produce explainable quality assessments aligned with radiologist diagnostic criteria. Integrating CT-MUSIQ's visual features with a domain-adapted language model to generate textual quality reports alongside numerical scores would significantly enhance clinical interpretability.

**6. Prospective clinical deployment study.** Embedding CT-MUSIQ within a hospital PACS pipeline and conducting a prospective study measuring its agreement with radiologist quality flags, its scan rejection accuracy, and its correlation with diagnostic outcomes would provide the clinical validation necessary for regulatory approval and widespread adoption.

**7. Explainability through attention visualisation.** CT-MUSIQ's Transformer self-attention maps could be visualised to identify the image regions and patches that most strongly influence the quality score prediction, providing a degree of explainability that may help radiologists understand and trust the automated assessment.

**8. Extension to volumetric quality assessment.** The current work operates on individual 2D axial slices. A volumetric extension that aggregates quality scores across all slices of a CT volume—or processes slices jointly through a temporal/volumetric attention mechanism—would better reflect clinical practice, where overall scan quality is assessed at the volume level.

**9. Self-supervised pre-training on unlabelled CT.** Given the chronic shortage of annotated CT quality data, self-supervised pre-training strategies adapted for CT images (e.g., masked image modelling in HU space) could provide stronger CT-domain representations as the initialisation for CT-MUSIQ, potentially surpassing the ImageNet-pretrained ViT-B/32 starting point.

## 6.5 Broader Impact and Ethical Considerations

The deployment of automated AI systems in clinical medical imaging carries significant ethical responsibilities that this section briefly examines, recognising that the thesis work itself is an early research contribution rather than a clinically deployed system.

### 6.5.1 Radiation Safety and the ALARA Principle

CT-MUSIQ is motivated in part by the goal of supporting the ALARA principle: by providing reliable automated quality assessment, the system could enable real-time scan rejection and immediate re-acquisition at the dose level that achieves diagnostic quality, rather than systematically exposing patients to higher doses "to be safe." However, this goal introduces a dual risk: if CT-MUSIQ incorrectly accepts a poor-quality scan (false positive), a patient may receive a subdiagnostic examination that delays care; if it incorrectly rejects a good-quality scan (false negative), the patient receives unnecessary additional radiation. The appropriate operating threshold for a clinical deployment must be determined through careful clinical validation studies that explicitly quantify the trade-off between these two error types.

### 6.5.2 Bias and Fairness

The LDCTIQAC 2023 dataset, while a valuable benchmark, was collected under specific acquisition conditions and annotated by a specific pool of radiologists. The model trained on this data may inherit biases from the annotation process, including systematic biases related to the specific scanner parameters used, the demographics of the patient cohort, or the preferences of the annotating radiologists. Before clinical deployment, it would be essential to evaluate CT-MUSIQ on data from diverse patient populations, scanners, and annotation teams to assess the robustness and fairness of its quality predictions across different subgroups.

### 6.5.3 Transparency and Explainability

A clinically deployed AI system must be explainable to the radiologists and radiographers who interact with it. CT-MUSIQ, as a black-box neural network, does not currently provide any explanation for its quality score predictions beyond the numerical score itself. Attention visualisation techniques—such as attention rollout, Grad-CAM applied to the `[CLS]` token, or attention map averaging across layers—could provide heatmaps indicating which image regions most strongly influenced the quality prediction. Such visualisations would help radiographers understand why a scan was flagged as low quality, potentially identifying specific artefacts that the model found problematic. Developing and validating these explainability tools is an important step toward clinical trust and regulatory approval.

### 6.5.4 Regulatory Pathway

In most jurisdictions, AI-based clinical decision support tools (including automated image quality assessment systems) require regulatory approval before clinical deployment. In the European Union, such systems are classified as medical devices under the MDR 2017/745 and require conformity assessment by a notified body. In the United States, FDA clearance is required under the 510(k) pathway for software as a medical device (SaMD). These regulatory processes require extensive clinical validation evidence, cybersecurity assessment, and quality

management system certification. CT-MUSIQ, as a research prototype, is far from meeting these requirements, but the work presented in this thesis provides a scientifically rigorous foundation upon which a regulatory-grade system could be built.

## 6.6 Reflection on the Research Process

This thesis was developed alongside the implementation of the CT-MUSIQ codebase, creating a close integration between experimental work and documentation. Several methodological lessons emerged that are worth recording for future researchers.

**Dataset inspection.** An early challenge was discovering that the dataset contained brain CT slices rather than the abdominal CT implied by the original challenge description. This finding—made through careful inspection of the raw images—led to the critical methodological adaptation of using brain soft-tissue windowing, which measurably improved model performance. Thorough dataset inspection is a prerequisite for any medical AI project, and domain expertise is essential to recognising when reported dataset metadata does not match the actual image content.

**Centralised configuration.** Maintaining a single `config.py` file as the sole repository of all hyperparameters proved invaluable during the iterative training and ablation process. When a hyperparameter needed adjustment, a single change propagated correctly to all scripts, eliminating the risk of inconsistencies between training and evaluation configurations. This practice is strongly recommended for all machine learning research projects involving multiple interacting scripts.

**Training monitoring.** The training logs revealed unexpected patterns in the validation aggregate that prompted investigation of the stochastic training dynamics. Rather than being a cause for concern, these patterns were identified as normal behaviour of gradient descent on a small dataset. Systematic monitoring of training dynamics through detailed logging is essential for diagnosing model behaviour and distinguishing normal stochasticity from genuine training pathologies.

## 6.7 Closing Remarks

This thesis demonstrates that the challenge of automated perceptual quality assessment for brain low-dose CT imaging can be meaningfully addressed through a carefully designed combination of multi-scale Vision Transformer architecture, clinical domain adaptation, and principled regularisation. CT-MUSIQ achieves competitive state-of-the-art performance on the LDCTIQAC 2023 benchmark while operating entirely within the constraints of consumer GPU hardware, making its principles accessible to researchers without access to high-end computing infrastructure.

The work also reveals that ensemble methods combining architecturally diverse backbones can surpass even well-regularised single models on small medical datasets, a finding with broad implications for the development of deep learning systems in data-scarce clinical domains. The complementary nature of CT-MUSIQ and the Agaldran Combo baseline suggests that hybrid strategies combining multi-scale patch tokenisation with diverse convolutional-hierarchical features represent a particularly promising direction for future research.

Ultimately, the goal of this research is to contribute a step toward a future where every CT acquisition is automatically and reliably assessed for diagnostic quality, enabling real-time quality feedback to radiographers, objective benchmarking of reconstruction algorithms, and—most importantly—improved patient care through the consistent delivery of diagnostically acceptable images at the lowest achievable radiation dose.

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# Declaration

I hereby declare that the academic thesis that I submit is the research work and results that I conducted and achieved under the guidance of my adviser. To my knowledge, except where especially noted and acknowledged, this thesis does not contain any research findings that have been published or written by others, or any materials that have been used to obtain degrees or certificates from any other educational institution. Any contribution made by others to this research has been clearly illustrated and acknowledged in this thesis.

I also hereby declare that the results of this academic thesis are achieved under the guidance of my adviser during my study at this institution and that they are owned by the university.

All experiments described in this thesis were conducted using the LDCTIQAC 2023 challenge dataset under the terms of its public research license. The computational implementation, architectural design, training procedures, and experimental analyses presented herein are original contributions of the author, except where explicitly credited to prior works.

**Thesis Author (Signature):** _____

**Thesis Adviser (Signature):** _____

**Date:** _____

(mm / dd / yyyy)

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*April 2026*

# A Supplementary Material

This appendix provides supplementary technical material supporting the main thesis, including the full training log for the primary CT-MUSIQ run, detailed pseudocode for the training loop, an extended discussion of the hash-based positional encoding, and implementation notes for reproducing all experiments.

## A.1 Full Training Log Summary

Table A.1 presents epoch-by-epoch training log data for the primary CT-MUSIQ 100-epoch run. The table shows the learning rate, training loss, validation PLCC, SROCC, KROCC, and Aggregate at each selected epoch. The full log is available in `results/ct_musiq/ct_musiq_training_log`

**Table A.1.** Selected epochs from the CT-MUSIQ primary training log. Stage 1 (epochs 1–5): frozen encoder. Stage 2 (epochs 6–100): full fine-tuning. Best val. aggregate checkpoint is highlighted.

| Epoch | Stage | LR | Train Loss | PLCC | SROCC | Aggregate |
|---|---|---|---|---|---|---|
| 1 | 1 | $3.0 \times 10^{-4}$ | 0.877 | 0.877 | 0.901 | 2.513 |
| 2 | 1 | $3.0 \times 10^{-4}$ | 0.432 | 0.881 | 0.917 | 2.559 |
| 3 | 1 | $3.0 \times 10^{-4}$ | 0.360 | 0.884 | 0.906 | 2.534 |
| 4 | 1 | $3.0 \times 10^{-4}$ | 0.315 | 0.951 | 0.953 | 2.733 |
| 5 | 1 | $3.0 \times 10^{-4}$ | 0.343 | 0.886 | 0.926 | 2.591 |
| 6 | 2 | $5.0 \times 10^{-5}$ | 0.346 | 0.947 | 0.949 | 2.714 |
| 7 | 2 | $5.0 \times 10^{-5}$ | 0.269 | 0.939 | 0.947 | 2.695 |
| 8 | 2 | $5.0 \times 10^{-5}$ | 0.281 | 0.940 | 0.942 | 2.688 |
| 9 | 2 | $\sim 4.8 \times 10^{-5}$ | 0.272 | 0.957 | 0.962 | 2.765 |
| 10 | 2 | $\sim 4.7 \times 10^{-5}$ | 0.261 | 0.943 | 0.941 | 2.675 |
| 20 | 2 | $\sim 4.0 \times 10^{-5}$ | ∼0.22 | ∼0.95 | ∼0.95 | ∼2.70 |
| 50 | 2 | $\sim 2.5 \times 10^{-5}$ | ∼0.16 | ∼0.96 | ∼0.96 | ∼2.75 |
| 100 | 2 | $\sim 1.0 \times 10^{-6}$ | ∼0.12 | — | — | — |

**Observation:** The validation aggregate exhibits non-monotonic behaviour within Stage 1 (peak at epoch 4, dip at epoch 5), which is expected: stochastic gradient descent on a randomly initialised regression head can achieve transient high validation correlation before the head's weights overfit the training set. The best Stage 1 epoch (epoch 4, Aggregate = 2.733) is comparable to the Stage 2 early performance, indicating that the pretrained ViT-B/32 encoder already contains substantial quality-relevant representations without any medical domain fine-tuning.

## A.2 Training Loop Pseudocode

Algorithm **??** presents the pseudocode for the CT-MUSIQ two-stage training loop.

The pseudocode for the CT-MUSIQ two-stage training loop proceeds as follows. The model weights $\theta$ are initialised from pretrained ViT-B/32. EMA shadow weights $\theta_{\text{EMA}}$ are set to $\theta$. An AdamW optimiser with weight decay 0.01 and a GradScaler for AMP are initialised. Then, for each epoch from 1 to 100: (1) If epoch $\leq 5$ (Stage 1), the transformer encoder and CLS token are frozen and learning rate is set to $3 \times 10^{-4}$. If epoch $= 6$, all parameters are unfrozen and cosine LR scheduling is initialised. (2) For each training batch $(\mathbf{X}, \mathbf{q})$: forward pass in fp16 autocast, compute composite loss $\ell$, scale and backpropagate, unscale gradients, clip gradient norm to 1.0, step optimiser, update GradScaler, and update EMA weights: $\theta_{\text{EMA}} \leftarrow 0.999 \cdot \theta_{\text{EMA}} + 0.001 \cdot \theta$. (3) Validation with EMA weights: temporarily substitute $\theta_{\text{EMA}}$ for $\theta$, compute (PLCC, SROCC, KROCC, Aggregate) on the validation set, restore $\theta$. (4) If Aggregate improves, save checkpoint and reset patience counter; otherwise increment counter and break if counter reaches 20 (early stopping). (5) Step cosine LR scheduler for epoch $\geq 9$. The function returns the best saved checkpoint.

## A.3 Hash-Based Positional Encoding: Extended Analysis

The hash-based positional encoding described in Section 3.3 is a key architectural contribution of CT-MUSIQ. This section provides an extended analysis of its properties and compares it with alternative positional encoding strategies.

### A.3.1 Comparison with Standard Sinusoidal Encoding

Standard sinusoidal positional encodings assign a unique, deterministic vector to each position index $n$ using:

$$\text{PE}(n, 2k) = \sin\left(\frac{n}{10000^{2k/d}}\right) \tag{A.1}$$

$$\text{PE}(n, 2k+1) = \cos\left(\frac{n}{10000^{2k/d}}\right) \tag{A.2}$$

These encodings are well-suited for fixed-length sequences but cannot naturally encode the two-dimensional spatial structure (row, column) or the scale identity of multi-scale patch sequences. Naively assigning sequential position indices $0, 1, \ldots, 192$ to the 193 tokens does not convey any information about whether a token is at scale 0 or scale 1, or what its row/column position is within its respective grid.

### A.3.2 Comparison with 2D Learned Position Encodings

Standard learned 2D positional encodings assign an independent embedding vector to each $(row, column)$ position pair. For a fixed grid of size $G \times G$, this requires $G^2$ embedding vectors. For our two-scale configuration:

- Scale 0 ($7 \times 7$ grid): 49 positions
- Scale 1 ($12 \times 12$ grid): 144 positions
- Total: $49 + 144 = 193$ unique positions

However, this approach requires a separate embedding for each distinct $(scale, row, column)$ triple, totalling $193 \times 768 = 148{,}224$ parameters. More importantly, positions that share the same row or column index across different scales cannot share any embedding components, preventing the model from learning that "row 3 at scale 0" and "row 3 at scale 1" occupy a similar vertical position in the image.

### A.3.3 Advantages of the Hash-Based Factored Encoding

The hash-based encoding addresses both limitations through the additive factored decomposition in eq. (3.7). The key advantages are:

1. **Parameter efficiency**: $(2 + G_{\max} + G_{\max}) \times d = 28 \times 768 = 21{,}504$ parameters versus $193 \times 768 = 148{,}224$ for the full 2D lookup table ($7\times$ reduction).
2. **Generalisation to new scales**: If a third scale is added (e.g., $512 \times 512$, $G = 16$), only $G_{\max}$ needs to be updated from 13 to 17, adding $(17 - 13) \times 768 \times 2 = 6{,}144$ new embedding parameters rather than requiring a full retraining of the positional encoding.
3. **Row/column sharing across scales**: The row embedding at row 3 is shared between scale 0 and scale 1, allowing the model to learn that vertical position at row 3 carries similar spatial meaning regardless of scale. This cross-scale position sharing may improve the model's ability to correlate quality features across scales.
4. **Scale embedding for scale-specific biases**: The dedicated scale embedding $\mathbf{E}_{\text{scale}}(s)$ allows the model to learn scale-specific biases (e.g., the coarse scale patches may systematically contribute differently to quality assessment than the fine scale patches), which is not possible with purely positional encodings.

## A.4 Ablation Study Raw Data

Table A.2 presents the complete validation aggregate scores for all six ablation configurations across the 30 training epochs. Note that these are validation aggregate values, not test set values.

**Table A.2.** Validation aggregate across ablation configurations (30 epochs). Values shown at selected epochs. Best configuration highlighted.

| Config | Epoch 10 | Epoch 20 | Epoch 30 | Scales |
|---|---|---|---|---|
| A1 (224, no KL) | ∼2.18 | ∼2.23 | 2.249 | [224] |
| A2 (224+384, no KL) | ∼2.19 | ∼2.25 | 2.271 | [224, 384] |
| A3 (KL=0.05) | ∼2.21 | ∼2.26 | 2.289 | [224, 384] |
| **A4 (KL=0.10)** | ∼2.23 | ∼2.28 | **2.305** | [224, 384] |
| A5 (KL=0.20) | ∼2.22 | ∼2.27 | 2.281 | [224, 384] |
| A6 (3-scale, KL=0.10) | ∼2.20 | ∼2.26 | 2.296 | [224, 384, 512] |

## A.5 Model Architecture Summary Table

Table A.3 provides a concise summary of the CT-MUSIQ architecture for quick reference.

**Table A.3.** CT-MUSIQ architecture summary.

| Component | Specification |
|---|---|
| Input | $512 \times 512$ greyscale CT slice, float32 $\in [0,1]$ |
| Preprocessing | Brain window (L=40, W=80 HU), channel replication $\times 3$ |
| Scale 0 | Resize to $224 \times 224$, extract $7 \times 7 = 49$ patches |
| Scale 1 | Resize to $384 \times 384$, extract $12 \times 12 = 144$ patches |
| Patch size | $32 \times 32$ pixels |
| Total tokens | 193 patch tokens + 1 CLS token = 194 |
| Patch embedding | Conv2d(3, 768, 32, 32) $\rightarrow$ Flatten $\rightarrow \mathbb{R}^{768}$ |
| Positional encoding | $\mathbf{E}_{\text{scale}}(s) + \mathbf{E}_{\text{row}}(r) + \mathbf{E}_{\text{col}}(c)$ |
| Transformer | 12 layers, 8 heads, $d = 768$, FFN dim= 3072, dropout= 0.05 |
| Global head | Linear(768, 384) $\rightarrow$ GELU $\rightarrow$ Dropout $\rightarrow$ Linear(384, 1) |
| Per-scale heads | $2 \times$ [MeanPool $\rightarrow$ Linear(768, 384) $\rightarrow$ GELU $\rightarrow$ Linear(384, 1)] |
| Total parameters | $\approx 88.3$ million |
| Output | Scalar quality score $\hat{q} \in \mathbb{R}$ (clipped to $[0,4]$ post-hoc) |
| Loss | $\mathscr{L}_{\text{MSE}} + 0.10 \cdot \mathscr{L}_{\text{KL}} + 0.50 \cdot \mathscr{L}_{\text{rank}}$ |
| Optimiser | AdamW, weight decay $= 0.01$ |
| Stage 1 | 5 epochs, LR $= 3 \times 10^{-4}$, encoder frozen |
| Stage 2 | 95 epochs, LR: $5 \times 10^{-5} \rightarrow 10^{-6}$ (cosine) |
| Batch size | 10 |
| EMA | $\rho = 0.999$ |
| Precision | FP16/FP32 AMP |

## A.6 Dataset Statistics

The LDCTIQAC 2023 dataset contains 1,000 brain CT slices with radiologist Likert quality scores. The score distribution across the full dataset has the following approximate statistics based on the provided `train.json` labels:

- **Score range**: Continuous float in $[0, 4]$
- **Mean score**: The majority of images receive mid-range scores (2–3), reflecting the clinical design of the dataset to include a range of quality levels.
- **Distribution**: The score distribution is approximately unimodal, with a peak around scores 2.5–3.0, and tails extending toward 0 (severely degraded) and 4 (excellent quality) that are less represented.
- **Training split mean**: Similar distribution to the overall dataset.
- **Implication for evaluation**: The use of all three correlation metrics (PLCC, SROCC, KROCC) rather than a single metric ensures that performance is evaluated across the full quality range, not just the most common quality levels.

## A.7 Evaluation Script Usage

The following commands reproduce the primary evaluation results reported in this thesis:

**CT-MUSIQ evaluation (standard):**

```
python evaluate.py --model ct_musiq
```

**CT-MUSIQ evaluation with TTA:**

```
python evaluate.py --model ct_musiq --tta
```

**Agaldran Combo evaluation:**

```
python evaluate.py --model agaldran_combo
```

**Ablation study:**

```
python ablation.py
```

All results are written to CSV files in the `results/` directory and can be compared against the published LDCTIQAC 2023 leaderboard values embedded in the `ct_musiq_results.csv` and `agaldran_combo_results.csv` files.

# B Glossary of Terms

This appendix defines the key technical terms used throughout the thesis.

**ALARA** As Low As Reasonably Achievable. The governing principle for ionising radiation exposure management in clinical practice.

**AMP** Automatic Mixed Precision. A training strategy that uses fp16 precision for forward pass activations and fp32 for gradients and parameter updates.

**BRISQUE** Blind/Referenceless Image Spatial Quality Evaluator. A classical NR-IQA algorithm based on Natural Scene Statistics.

**CLS token** Classification token. A learnable embedding vector prepended to the patch token sequence in Vision Transformers, used to aggregate global information for downstream tasks.

**CT** Computed Tomography. A medical imaging modality using X-ray projections and computed reconstruction to produce cross-sectional anatomical images.

**DLIR** Deep Learning Image Reconstruction. CT reconstruction algorithms based on neural networks, including CNNs and diffusion models.

**EMA** Exponential Moving Average. A smoothed average of model weights used to reduce the stochastic variability of the final trained model.

**FBP** Filtered Back Projection. The classical analytical CT reconstruction algorithm.

**FR-IQA** Full-Reference Image Quality Assessment. Methods that evaluate image quality through comparison with a perfectly aligned reference image.

**HU** Hounsfield Unit. The quantitative scale of CT pixel values, where water is 0 HU and air is approximately $-1000$ HU.

**IQA** Image Quality Assessment. The task of evaluating the perceptual quality of an image, either through manual subjective assessment or automated algorithms.

**KL divergence** Kullback–Leibler divergence. A measure of how one probability distribution differs from another. Used in CT-MUSIQ as a scale-consistency loss.

**KROCC** Kendall Rank-Order Correlation Coefficient. A rank correlation metric measuring the proportion of concordant pairs among all pairs of predictions.

**LDCT** Low-Dose Computed Tomography. CT acquisition with reduced radiation dose, achieved by lowering the X-ray tube current.

**LDCTIQAC** Low-Dose Computed Tomography Perceptual Image Quality Assessment Grand Challenge. The MICCAI 2023 challenge that provided the benchmark dataset used in this thesis.

**Likert scale** A rating scale used in the radiologist annotation process, ranging from 0 (unac-

ceptable) to 4 (excellent) in the LDCTIQAC 2023 dataset.

**MHSA** Multi-Head Self-Attention. The core computational mechanism of Transformer models, computing attention-weighted summaries of the input sequence.

**MSE** Mean Squared Error. The primary regression loss function used in CT-MUSIQ.

**MUSIQ** Multi-Scale Image Quality Transformer. The natural image IQA model from Ke et al. (ICCV 2021) that serves as the architectural foundation for CT-MUSIQ.

**NPS** Noise Power Spectrum. A frequency-domain characterisation of the statistical properties of CT image noise.

**NR-IQA** No-Reference Image Quality Assessment. Methods that evaluate image quality without access to any reference image.

**NSS** Natural Scene Statistics. The statistical regularities exhibited by high-quality natural photographs, exploited by classical blind IQA algorithms.

**PACS** Picture Archiving and Communication System. The hospital information system used to store, retrieve, and display medical images.

**PLCC** Pearson Linear Correlation Coefficient. A metric measuring linear correlation between predicted and true quality scores.

**SROCC** Spearman Rank-Order Correlation Coefficient. A metric measuring the agreement of rank orderings between predicted and true quality scores.

**ViT** Vision Transformer. A class of image processing neural networks that replace convolutional layers with Transformer self-attention mechanisms.

**ViT-B/32** Vision Transformer Base model with patch size 32. The pretrained model whose weights are used to initialise the CT-MUSIQ encoder.

**VLM** Vision-Language Model. A large neural network that jointly processes visual and textual inputs for tasks requiring both modalities.